Mathematical Modeling of a Dynamic Environment: A Functional Approach to Quantifying Health State

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Abstract

This article proposes a formal mathematical framework for modeling a dynamic environment and its temporal evolution. We introduce a hierarchy of three fundamental objects: the frame, the events and the time axis.

Each event is characterized by five quantifiable attributes that the operator Γ extracts or computes, directly integrating the temporal dynamics into the expression of gravity. To capture the interactions between coexisting events (amplification, neutralization, feedback), we define an interaction matrix $\mathbf{W}(t)$.

The operator ζ synthesizes all of this information into a normalized scalar (the health state of the environment) with values in $(-1, 1)$, based on the tanh function. A preliminary experimental validation supports the improvements brought (dilation, adaptive splines, clustering) and identifies the current limits of the model.

Note on contributions.
The following elements constitute original contributions of the author: the functional form of Γ with temporal residual gravity, the per-cluster calibration procedure, the operator ζ_λ with dilation, the construction of the variational weight sheet and the strategy of nodes adaptive to discontinuities. The following results come from the literature: properties of tanh, the k-means algorithm [6], the Q-Learning rule [4], cubic splines [5], the Dirichlet functional [7].

*The author certifies that all of the mathematical reasoning, the formal model and the algorithmic constructions presented in this document are the result of original personal research unless otherwise stated.

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1 Introduction

Any attempt to model a real phenomenon begins with the need to formalize what is, at first sight, mere intuition. The environment of an individual or of a system is made up of events, structures and interactions that unfold in time and in space. Before building an operational model, it is appropriate to establish its conceptual and mathematical foundations.

1.1 Problem statement and objectives

The central objective is to model the health state of an environment (Definition 6.1): a synthetic quantity measuring its propensity to evolve positively or negatively at a given instant. Such a normalized indicator is of direct interest in any context of anticipating a complex system such as a financial market, an ecosystem or an industrial organization.

1.2 Complexity of the representation and chosen approach

The difficulty lies in the diversity of environments: some are tangible (a city, a forest), others intangible (a social network, a stock market). We adopt an event-based representation: the environment is entirely described by its constituent events, each characterized by five quantifiable attributes, without explicitly modeling the spatial frame.

1.3 Applications and stakes

- Finance: quantify the state of a market from macroeconomic events in order to anticipate the direction of an asset.
- Ecology: assess the resilience of an ecosystem to disturbances and estimate its tipping point.
- Risk management: synthesize operational incidents into a single indicator that is understandable and actionable.

1.4 Main contributions

- Mathematical formalization of a dynamic environment over $\tau = \mathbb{R}^+$.
- Operator Γ with temporal residual gravity, integrating the remaining duration directly into the expression of gravity.
- Interaction matrix $\mathbf{W}(t)$, formalizing the cross effects between coexisting events.
- Operator ζ with values in $(-1, 1)$, based on tanh.
- Variational weight sheet, making it possible to assign an initial weight to any new event by minimizing the normalized Dirichlet functional.

2 Framework and fundamental definitions

2.1 Basic mathematical objects

Definition 2.1 (Environment)**.** [Original contribution of the author.]
An environment is a pair $\mathcal{E} = \big(C, \{e_1, \dots, e_N\}\big)$*, where* C *denotes the frame and* $\{e_i\}_{i=1}^{N}$ *is the finite set of its events, with* $N \in \mathbb{N}^*$*.*

Remark 2.1. *Such a definition is meant to be static. The time dependence is introduced in Section 2.2.*

Definition 2.2 (Frame)**.** [Original contribution of the author.]
The frame C *is the arbitrary boundary that the user imposes on the study of the environment. It may be geographical, thematic or of any other nature. It is a subjective decision exogenous to the model.*

Definition 2.3 (Event)**.** [Original contribution of the author.]
An event is the smallest distinctly identifiable entity within the environment. Each event e *is characterized by five quantifiable attributes presented in Section 3.*

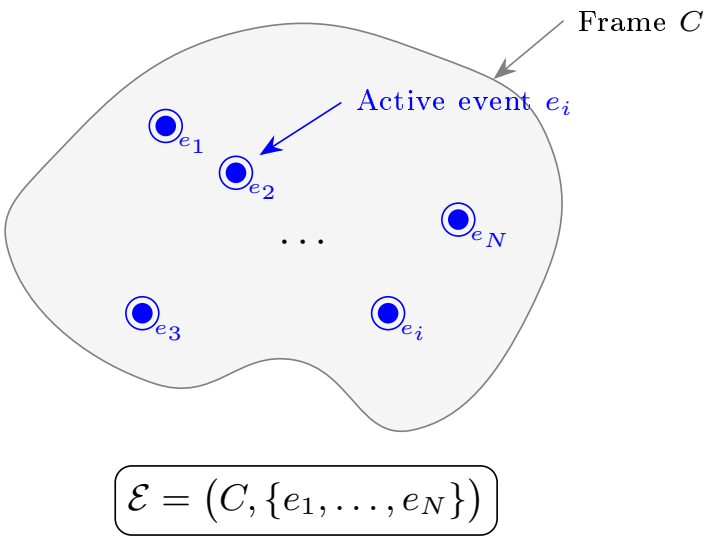


FIGURE 1 – Schematic representation of the environment $\mathcal{E}$

2.2 Time axis and dynamics

Definition 2.4 (Time axis)**.** *The time axis is defined by* $\tau = \mathbb{R}^+$*. All the quantities of the model are therefore considered as functions of time* $t \in \tau$*.*

However, since the set $\mathbb{R}^+$ is continuous, its computer representation and processing are delicate in practice. In order to make the model numerically exploitable, we introduce a discretization of time by replacing $\mathbb{R}^+$ with a discrete set of indexed instants:

$$\tau_d = \mathbb{N}.$$

Time is thus represented by a succession of integer indices corresponding to the instants of observation or update of the system.

Definition 2.5 (Dynamic environment)**.** [Original contribution of the author.]

$$\mathcal{E} : \tau \longrightarrow \mathbb{E}, \qquad t \longmapsto \mathcal{E}(t) = \Big(C(t), \big\{e_1, \dots, e_{N(t)}\big\}\Big).$$

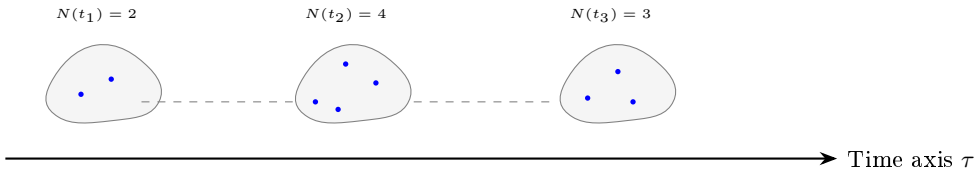


FIGURE 2 – Dynamic evolution of $\mathcal{E}(t)$

Remark 2.2. *$N(t)$ is assumed to be locally bounded on every compact $[0, T]$, guaranteeing that the finite sums used later are well defined.*

Remark 2.3 (Order-of-magnitude filtering)**.** *From a realistic implementation perspective, it is generally not possible to exhaustively consider all the events present in a given environment.*

Indeed, an event may give rise to consequences that themselves become new events, which in turn produce other effects. This recursive propagation quickly leads to a combinatorial growth of the number of potentially observable events. Thus, from a computational standpoint, it becomes necessary to introduce a filtering mechanism in order to limit the space of events actually taken into account in the model.

We then associate with the environment a characteristic order of magnitude denoted h, as well as an order of magnitude h' specific to each event. The idea is to retain only the events whose scale remains comparable to that of the environment considered.

In other words, only the events satisfying a condition of the type

$$h' \sim h$$

are kept in the analysis.

This choice is naturally justified:

- *events of too small an order of magnitude have a negligible influence on the overall dynamics;*
- *conversely, events of excessively high order of magnitude often belong to a dynamic external to the frame under study.*

Order-of-magnitude filtering thus makes it possible to keep a model that is computable, stable and consistent with the effective scale of the environment analyzed.

Remark 2.4 (Practical determination of h and h')**.** *This filtering mechanism is a recent addition to the formal framework, and its practical implementation deserves to be specified. h is not a quantity computed unambiguously by the model: like the frame C (Definition 2.2), it is an exogenous choice, fixed by the user according to the scale at which they wish to study the environment (for example, a "city" scale will by construction exclude events of "neighborhood" or "country" scale).*

On the other hand, the order of magnitude h' specific to an event can be approximated by a simple computation from its primary attributes, for example $h' \propto D \cdot F \cdot R$, a proxy for its spatio-temporal magnitude. Once h is fixed by the user, the condition $h' \sim h$ can therefore be evaluated automatically for each candidate event. This mechanism remains, at this stage, a pre-filtering heuristic rather than a calibrated component of the model; its complete automation (computation of h itself from the frame C) is left as a perspective (Section 9).

3 Characterization of events

3.1 The five attributes of an event

Symbol	Name	Domain	Description
D	Duration	$\mathbb{R}^+$	Time during which the event remains active.
F	Frequency	$\mathbb{R}^+$	Number of occurrences per unit of time.
R	Radius of action	$\mathbb{R}^+$	Spatial or thematic extent of the influence.
G	Gravity	$\mathbb{R}$	Intensity of the intrinsic impact of the event.
P	Predictability	$\mathbb{R}^+$	Degree to which the event can be anticipated.

Remark 3.1. *The gravity G can be interpreted as the resultant of the four other attributes.*

3.2 The characterization operator Γ

Definition 3.1 (Operator Ψ — general form)**.** [Original contribution of the author.]
Let $\mathbb{A}$ be the space of character strings:

$$\Psi : \mathbb{A} \longrightarrow \mathbb{R}^5, \qquad e \longmapsto \big(D, F, R, G, P\big). \tag{1}$$

In a practical implementation, Ψ can be realized by a natural language processing model. The mathematical definition remains independent of the chosen implementation.

The operator Γ introduced below (Definition 3.2) corresponds to the restriction of Ψ to the sole computation of gravity from the four primary attributes.

Definition 3.2 (Operator Γ — reduced form)**.** *When the event is represented by its four primary attributes:*

$$\Gamma : \mathbb{R}^4 \longrightarrow \mathbb{R}, \qquad (D, F, R, P) \longmapsto G. \tag{2}$$

Definition 3.3 (Functional form of Γ)**.** [Original contribution of the author.]

$$G = \Gamma(\mathcal{D}, \mathcal{F}, \mathcal{R}, \mathcal{P}) = \frac{\mathcal{D}^{\alpha} \cdot \mathcal{F}^{\beta} \cdot \mathcal{R}^{\delta}}{1 + \mathcal{P}^{\gamma}}, \tag{3}$$

where $\alpha, \beta, \delta, \gamma > 0$ are parameters to be calibrated. Gravity increases with duration, frequency and radius of action, and decreases with predictability (predictable events can be mitigated upstream). This is a modeling choice; the monotonicity and decrease properties that follow from it are established in Annex 9.4.

3.3 Integration of the temporal dynamics into Γ

Definition 3.4 (Temporal residual gravity)**.** [Original contribution of the author.]
Let an event e appear at t_0 with a total duration $\mathcal{D}$. Its effective duration at time t is $\mathcal{D}_{\text{eff}}(t) = \max(\mathcal{D} - (t - t_0), 0)$. The time-dependent gravity is:

$$\mathcal{G}(t) = \Gamma\big(\mathcal{D}_{\text{eff}}(t), \mathcal{F}, \mathcal{R}, \mathcal{P}\big) = \frac{\big(\max(\mathcal{D} - (t - t_0), 0)\big)^{\alpha} \cdot \mathcal{F}^{\beta} \cdot \mathcal{R}^{\delta}}{1 + \mathcal{P}^{\gamma}}. \tag{4}$$

Remark 3.2. *As soon as $t \geq t_0 + \mathcal{D}$, gravity vanishes exactly: the event no longer influences the health state. This formulation is consistent with the adoption of static weights, since the temporal extinction of the influence is entirely driven by $\mathcal{D}_{\text{eff}}(t)$ in Γ.*

3.4 Calibration of hyper-parameters by cluster of events

3.4.1 Introduction

A single set of parameters $(\alpha, \beta, \delta, \gamma)$ presupposes a structural homogeneity of all events. In practice, however, events group into families with distinct gravity laws.

Remark 3.3 (Gravity laws)**.** *By gravity law of a family of events, we mean the specific functional relation linking $(\mathcal{D}, \mathcal{F}, \mathcal{R}, \mathcal{P})$ to the observed real impact — entirely characterized by the vector of exponents $(\alpha, \beta, \delta, \gamma)$.*

Two families have different laws if their optimal exponents differ. For example, for an event that is:

— *Financial: the duration $\mathcal{D}$ tends to dominate (α high).*

— *Health-related: the frequency $\mathcal{F}$ and the radius $\mathcal{R}$ are predominant (β, δ high).*

Remark 3.4 (Empirical justification of the cluster approach)**.** *The theoretical argument above (expected heterogeneity of gravity laws) motivates the choice, adopted from this section on and kept throughout the article, of calibrating Γ locally by cluster rather than globally. This choice is also confirmed a posteriori by experiment: the quantitative comparison between a global model (single set of parameters) and a per-cluster model, presented in Section 7.4, shows a systematic gain in accuracy in favor of the latter approach. The reader in a hurry to see this numerical confirmation may refer to it directly; we keep here the pedagogical order of presentation (construction of the per-cluster model, then validation) rather than a premature validation before all the objects are defined.*

Definition 3.5 (Partition into clusters)**.** [Original contribution of the author.]
Let $\mathcal{B} = \{e_1, \ldots, e_N\}$ be the event base. A partition into K clusters is $\mathcal{B} = C_1 \sqcup \cdots \sqcup C_K$ such that each C_k groups events with similar profiles $(\mathcal{D}, \mathcal{F}, \mathcal{R}, \mathcal{P})$.

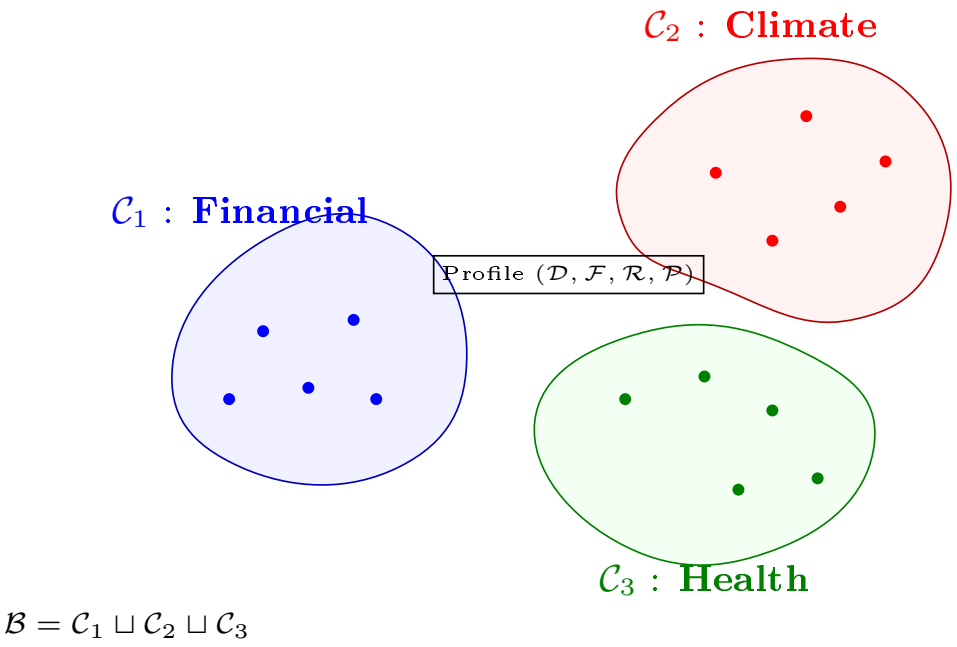


FIGURE 3 – Partition of $\mathcal{B}$ into disjoint clusters

The potato-shaped forms represent regions of similarity in the space of events, within which the parameters $(\alpha_k, \beta_k, \delta_k, \gamma_k)$ are calibrated independently.

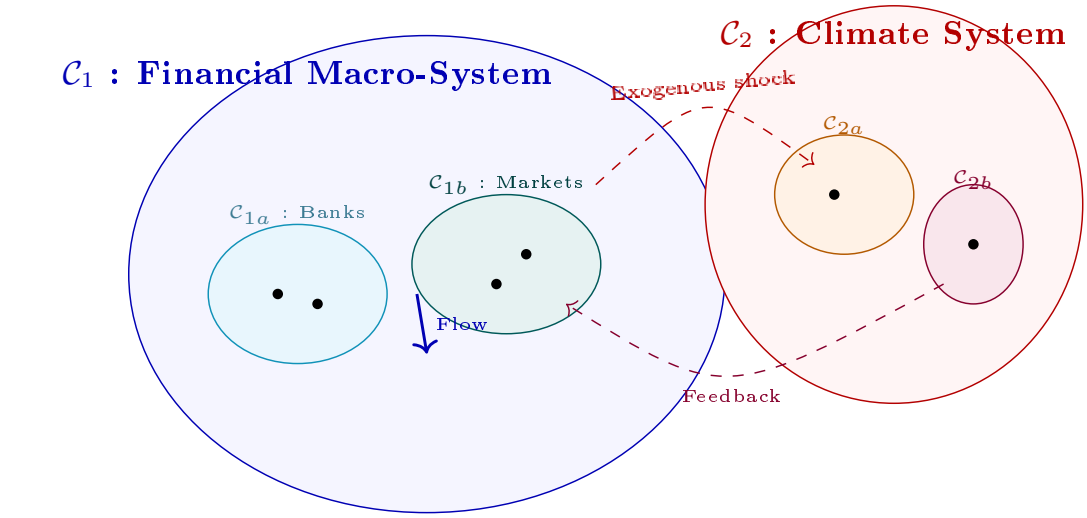


FIGURE 4 – Hierarchical and synaptic modeling of $\mathcal{B}$.

The macro-clusters ($\mathcal{C}_1$, $\mathcal{C}_2$) encompass specialized substructures. The nodes there represent isolated events (comparable to neurons), which interact through local influence parameters (internal synapses) and systemic shocks (inter-cluster axons).

Ultimately, this organization into clusters proves essential for the design of our mathematical model: it makes it possible to target specific events and to efficiently categorize those exhibiting strong similarities.

Definition 3.6 (MAPE)**.** [Standard result from the literature.]
The Mean Absolute Percentage Error measures the mean relative deviation:

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \ (\%). \tag{5}$$

A MAPE of 0% is a perfect prediction; below 5%, it is considered very satisfactory in risk modeling.

Definition 3.7 (Per-cluster calibration)**.** [Original contribution of the author.]
For each cluster C_k, we learn $\theta_k = (\alpha_k, \beta_k, \delta_k, \gamma_k)$ by minimizing the MAPE:

$$\hat{\theta}_k = \arg\min_{\theta \in \mathbb{R}^4_{>0}} \frac{1}{|C_k|} \sum_{e_i \in C_k} \frac{|\Gamma_\theta(e_i) - \mathcal{G}_i^{\text{db}}|}{\mathcal{G}_i^{\text{db}}}, \tag{6}$$

where $\mathcal{G}_i^{\text{db}} \in \mathbb{R}^+$ is the reference gravity of e_i, coming from a historical database or an expert assessment — the real impact observed a posteriori, as opposed to the predicted gravity $\Gamma_\theta(e_i)$ computed a priori. The pair $(\hat{\theta}_k, k(i))$ thus defined — the cluster's parameter vector, and the index of the cluster containing e_i — is reused as is in the rest of the article; we will briefly recall it at each new occurrence far from this definition.

The operator Γ becomes locally adapted:

$$\Gamma(e_i) = \Gamma_{\hat{\theta}_{k(i)}}(e_i) \tag{7}$$

where $k(i)$ denotes the cluster of e_i.

In order to take into account the heterogeneity of the events present in the database, we introduce a global formulation of the operator Γ based on a local calibration per cluster.

The idea is to associate with each cluster C_k a specific operator, calibrated independently on the data belonging to this cluster. The global operator is then obtained by aggregating these local operators.

Definition 3.8 (Global operator by cluster aggregation)**.** *For each cluster C_k, we define a local operator:*

$$\Gamma_k := \Gamma_{\hat{\theta}_k}, \tag{8}$$

where $\hat{\theta}_k$ is obtained by the calibration procedure introduced in Definition 3.7.

The global operator Γ is then written in the form:

$$\Gamma(e_i) = \sum_{k=1}^{K} \mathbb{1}_{\{e_i \in C_k\}} \Gamma_k(e_i), \tag{9}$$

where:

— *K denotes the total number of clusters.*

— *$\mathbb{1}_{\{e_i \in C_k\}}$ is the indicator function equal to 1 if $e_i \in C_k$, and 0 otherwise.*

This writing amounts to selecting the local operator associated with the cluster to which the event e_i belongs. Thus, we obtain the equivalent form:

$$\Gamma(e_i) = \Gamma_{k(i)}(e_i), \tag{10}$$

where $k(i)$ denotes the index of the cluster containing the event e_i.

Remark 3.5 (Advantages of per-cluster calibration)**.** *This approach has several important advantages:*

— *Flexibility: each cluster has its own parameters $\hat{\theta}_k = (\alpha_k, \beta_k, \delta_k, \gamma_k)$, which makes it possible to locally adapt the behavior of the operator Γ to the specific characteristics of each family of events.*

— *Interpretability: the learned parameters keep a physical or empirical meaning. For example, a high value of α_k reflects a strong influence of duration in the evaluation of the gravity of the events of the cluster considered.*

— *Robustness: the separation into clusters limits the influence of atypical behaviors between different categories of events, which improves the overall stability of the calibration.*

— *Extensibility: adding a new type of event does not require retraining the whole global model. It suffices to learn a new set of parameters $\hat{\theta}_{K+1}$ associated with the corresponding cluster.*

— *Modularity: each local operator Γ_k can be studied, modified or improved independently of the other clusters, which facilitates future updates of the model.*

— *Better local accuracy: certain families of events exhibit very different dynamics. A specific calibration per cluster therefore yields a better local approximation than a single global model.*

3.5 Lexical construction of clusters

3.5.1 Introduction

We have studied how the introduction of clusters could improve our model. However, we have not yet looked into how to construct these clusters. To this end, we propose a lexical approach, in which events are grouped according to the keywords they share.

3.5.2 Development

The general idea is that two events with a similar vocabulary will be considered close in the representation space.

Consider the following event:

"A large-scale fire broke out in the Amazon during a drought."

In this description, the terms *fire* and *Amazon* make it possible to associate the event with the category of forest fires. The term *drought* can, for its part, be interpreted as a secondary event or an independent aggravating factor.

Finally, the expression *large-scale* corresponds more to a quantitative characterization of the event than to a lexical category; this intensity will therefore be taken into account directly by our gravity model.

Proposition 3.1 (Lexical clustering of events)**.** [Original contribution of the author.]
Let an event e_i be described by a set of keywords:

$$e_i = \{w_{i1}, w_{i2}, \ldots, w_{in}\}. \tag{11}$$

The proximity between two events e_i and e_j can be evaluated by a lexical similarity measure defined by:

$$\mathrm{Sim}(e_i, e_j) = \frac{|e_i \cap e_j|}{|e_i \cup e_j|}. \tag{12}$$

Two events are then associated with the same lexical cluster when their similarity exceeds a fixed threshold:

$$\mathrm{Sim}(e_i, e_j) \geq \tau, \tag{13}$$

where $\tau \in [0, 1]$ denotes the grouping threshold.

This method automatically generates categories of events based on their lexical similarities, while guaranteeing groupings (clusters) whose interpretation remains intuitive. It thus becomes possible to progressively refine the granularity of the classification, in order to create ever more specific and targeted subfamilies of events.

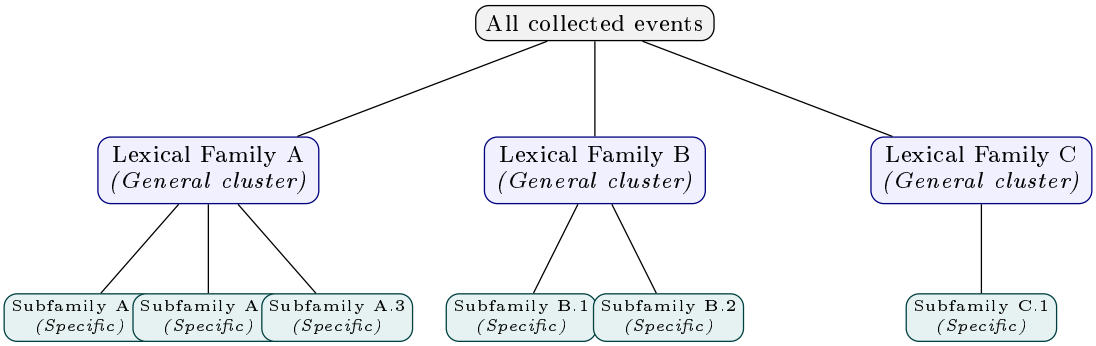


FIGURE 5 – Grouping of events into successive subfamilies

3.6 Reformulation of the dynamic environment

We substitute each event by its image under Γ with the calibrated parameters $\hat{\theta}_{k(i)}$ of Definition 3.7:

$$\mathcal{E}(t) = \Big(C(t), \big\{\Gamma_{\hat{\theta}_{k(1)}}(e_1), \Gamma_{\hat{\theta}_{k(2)}}(e_2), \ldots, \Gamma_{\hat{\theta}_{k(N(t))}}(e_{N(t)})\big\}\Big), \quad \forall\, t \in \tau. \tag{14}$$

Remark 3.6. *The index $\hat{\theta}_{k(i)}$ in the equation above is essential: it means that each event is characterized by the parameters of its own cluster, and not by a global set. This dependence on θ_k is an original contribution of this work.*

4 Interactions between events

4.1 Types of interactions

Definition 4.1 (Types of interactions)**.** [Original contribution of the author.]
Let e_i and e_j be two coexisting events. We distinguish:

— *Amplification: e_j reinforces the impact of e_i (e.g., drought → fire).*

— *Feedback: e_i and e_j sustain each other mutually.*

— *Neutralization: e_j attenuates the impact of e_i (e.g., preventive intervention → natural disaster).*

4.2 Interaction matrix

Definition 4.2 (Interaction matrix)**.** [Original contribution of the author.]
The interaction matrix $\mathbf{W}(t) \in \mathbb{R}^{N(t)\times N(t)}$ encodes the type of interaction (Definition 4.1) of e_j on e_i:

— *$W_{ij}(t) > 0$: amplification or feedback — e_j reinforces e_i.*

— *$W_{ij}(t) < 0$: neutralization — e_j attenuates e_i.*

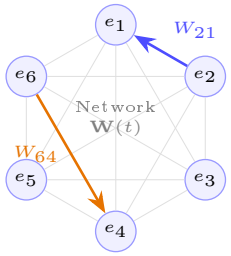
— *$W_{ij}(t) = 0$: no interaction between e_i and e_j.*

Remark 4.1 (Amplification / feedback distinction)**.** *The sign of $W_{ij}(t)$ alone does not make it possible to distinguish a simple amplification from a true feedback (Definition 4.1).*

We propose to formalize this distinction as follows:

— *The interaction of e_j on e_i is a simple amplification when $W_{ij}(t) > 0$ and $W_{ji}(t) = 0$ (one-way influence).*

— *The interaction of e_j on e_i is a feedback when $W_{ij}(t)\,W_{ji}(t) > 0$ (the two events sustain each other mutually).*



$$\longrightarrow \quad \mathbf{W}(t) = \begin{pmatrix} 0 & W_{12}(t) & \cdots & W_{1N}(t) \\ W_{21}(t) & 0 & \cdots & W_{2N}(t) \\ \vdots & \vdots & \ddots & \vdots \\ W_{N1}(t) & W_{N2}(t) & \cdots & 0 \end{pmatrix}$$

FIGURE 6 – Correspondence between interaction network and matrix representation

Since we have introduced the notion of influence, gravity can no longer be computed simply from its local parameters alone, but must also take into account those of the neighboring events.

This is why we introduce a new form of the gravity law, integrating the interactions between events.

Definition 4.3 (Effective gravity)**.** [Original contribution of the author.]

$$\widetilde{\mathcal{G}}_i(t) = \mathcal{G}_i(t) + \sum_{\substack{j=1 \\ j\neq i}}^{N(t)} W_{ij}(t)\,\mathcal{G}_j(t). \tag{15}$$

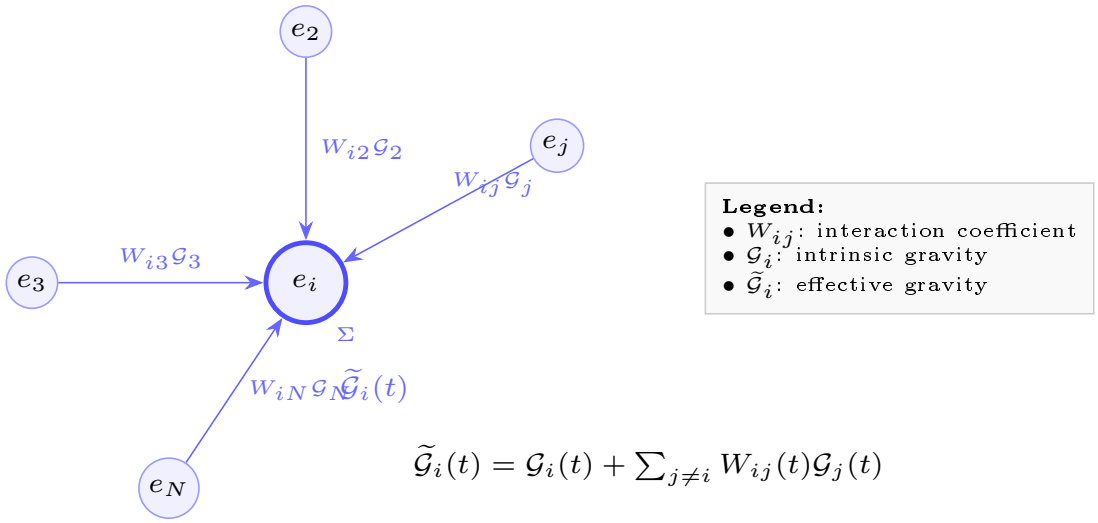


FIGURE 7 – Computation of $\widetilde{\mathcal{G}}_i$

For implementation purposes, vector and matrix formulations are preferred, as they are both simpler to manipulate and more efficient than explicit loops in Python.

We therefore translate the additive form of gravity into a compact matrix writing.

Proposition 4.1 (Vector form of effective gravity)**.** *[3]*
We define the effective gravity by:

$$\widetilde{\mathcal{G}}(t) = \big(\mathbf{I} + \mathbf{W}(t)\big)\,\mathcal{G}(t).$$

Setting the augmented influence matrix:

$$\mathbf{W}^{\star}(t) := \mathbf{I} + \mathbf{W}(t),$$

we obtain an equivalent more compact writing:

$$\widetilde{\mathcal{G}}(t) = \mathbf{W}^{\star}(t)\,\mathcal{G}(t),$$

where $\mathbf{W}^{\star}(t)$ directly integrates the own effects (diagonal) and the interactions between events (off-diagonal).

Proof. See Annex 9.4. □

5 Algorithmic framework

Algorithm 1 Computation of $\zeta_\lambda(\mathcal{E}(t))$

Input: $\{e_i\}$, $\mathbf{W}(t)$, $\{\omega_i\}$, $\lambda > 0$.
Step 1 - Gravities:
for $i = 1$ **to** $N(t)$ **do**
    $\mathcal{G}_i(t) \leftarrow \Gamma_{\hat{\theta}_{k(i)}}(\mathcal{D}_{\text{eff},i}(t), \mathcal{F}_i, \mathcal{R}_i, \mathcal{P}_i)$ [eq. (4)]
end for
Step 2 - Effective gravities:
$\widetilde{\mathcal{G}}(t) \leftarrow \mathbf{W}^\star(t)\mathbf{G}(t)$
Step 3 - Aggregation with normalized static weights:
$\bar{\omega}_i \leftarrow \omega_i / \sum_j |\omega_j|$
$S(t) \leftarrow \sum_i \bar{\omega}_i \, \widetilde{\mathcal{G}}_i(t)$
Step 4 - Normalization:
$\zeta_\lambda(\mathcal{E}(t)) \leftarrow \tanh(S(t)/\lambda)$
Output: $\zeta_\lambda(\mathcal{E}(t)) \in (-1, 1)$

Remark 5.1. *Complexity dominated by Step 2:* $O(N(t)^2)$ *in a naive implementation.*

6 The health state of an environment

6.1 Introduction

This section addresses the core of the article's contribution. As a reminder, the objective is to build a measurable quantity making it possible to assess whether the environment is conducive, or not, to the evolution of the user.

All of the previous developments (ranging from the mathematical translation of the environment via its fundamental elements (the events, Section 3) to the modeling of the complexity of their interactions (Section 4)) constitute a solid basis for defining this quantification.

This quantity will be called the *health state*.

6.2 Motivation and definition

Definition 6.1 (Health state)**.** [Original contribution of the author.]
The health state $\zeta(\mathcal{E}(t)) \in (-1,1)$ *is a scalar function that quantifies, at each instant* t*, the overall propensity of the environment* $\mathcal{E}(t)$ *to induce a positive or negative evolution of the user.*

This quantity is normalized in the interval $(-1,1)$ in order to guarantee a uniform interpretation, independent of the scale of the events considered. This normalization can be interpreted as a generalization of the notion of percentage, where -1 corresponds to an entirely unfavorable situation (-100%) and $+1$ to an entirely favorable situation ($+100\%$).

More precisely:

— when $\zeta(\mathcal{E}(t)) \to +1$, the environment is strongly favorable, and the observed dynamics contribute mostly to an improvement of the user's state;

— when $\zeta(\mathcal{E}(t)) \to -1$, the environment is strongly unfavorable, reflecting an accumulation of negative or degrading events;

— when $\zeta(\mathcal{E}(t)) \approx 0$, the environment is neutral, the positive and negative effects offsetting each other overall.

Thus, $\zeta(\mathcal{E}(t))$ provides a normalized synthesis of the overall impact of the environment on the user.

Remark 6.1. *Of course, this notion of evolution depends intrinsically on the user considered. An environment may be hostile for a given individual while being globally favorable for another.*

We rely here on the idea of a balance between "good" and "evil", a concept widely evoked in various fields, notably religious, economic and philosophical writings.

6.3 The operator ζ

Definition 6.2 (Operator ζ)**.** [Original contribution of the author.]
The operator ζ *is defined by the mapping:*

$$\zeta : \mathbb{E} \longrightarrow (-1,1), \qquad \mathcal{E} \longmapsto \zeta(\mathcal{E}).$$

Evaluated at an instant t*, the operator is expressed by the function:*

$$\zeta(\mathcal{E}(t)) = \tanh\big(S(t)\big) = \tanh\left(\sum_{i=1}^{N_{\text{events}}(t)} \text{weight}_i \ \text{EffectiveGravity}_i\right)$$

Interval	Nature	Interpretation
$(-1,0)$	Negative	Degradation from the user's point of view.
$\{0\}$	Neutral	Balance; no net change.
$(0,1)$	Positive	Improvement from the user's point of view.

6.4 Choice of tanh and associated properties

The tanh function is used as the central normalization function of our model in order to transform an unbounded quantity $S(t) \in \mathbb{R}$ into a bounded indicator $\zeta(t) \in (-1, 1)$, interpretable as a health state.

Proposition 6.1 (Fundamental properties of tanh)**.** [Standard result from the literature.]
[2] The function $\tanh : \mathbb{R} \to (-1, 1)$ *is a strictly increasing bijection satisfying the following properties:*

— $\zeta(t) \in (-1, 1)$ *for every finite value of* $S(t)$*;*

— *the transformation preserves the sign:* $\text{sign}(\tanh(S)) = \text{sign}(S)$*;*

— *there is no loss of structural information,* tanh *being injective and monotone;*

— *the saturation at* ± 1 *allows a bounded and stable interpretation of the extremes.*

Proof. See Annex 9.4. □

The key property of tanh lies in its progressive saturation. Indeed,

$$\tanh'(x) = 1 - \tanh^2(x) \longrightarrow 0 \quad \text{as } |x| \to +\infty.$$

Thus, reaching the limit values ± 1 would presuppose an absolutely favorable or unfavorable environment, which is not realistic in the complex systems studied here. The tanh function thus formalizes a principle of impossibility of absolute extremes.

Finally, the local linearity in the neighborhood of the origin plays a fundamental role:

$$\tanh(S) \approx S - \frac{S^3}{3} + O(S^5).$$

This property guarantees that ordinary situations ($S \approx 0$) undergo no artificial distortion, while ensuring a progressive transition toward extreme regimes.

6.5 Static weights and L^1 normalization

6.5.1 Introduction

The temporal decay of the influence of an event is entirely governed by the term $\mathcal{D}_{\text{eff}}(t)$ within the operator Γ (Section 3.3), via an exponential function controlled by the hyperparameter α.

In this context, it is not necessary for the weights associated with events to be dynamic, since the effective duration already integrates this temporal dimension. They are therefore considered structurally static: they do not vary explicitly as a function of time, but depend implicitly on the current state of the system.

6.5.2 Definition and interpretation of the weights

Definition 6.3 (Initial weight)**.** [Original contribution of the author.]
Each event e_i *is associated with a weight* $\omega_i \in [-\omega_{\max}, \omega_{\max}]$*, representing its intrinsic influence on the dynamics of the environment, according to the following convention:*

— $\omega_i > 0$*: favorable contribution, improving the overall state;*

— $\omega_i < 0$*: unfavorable contribution, degrading the overall state;*

— $|\omega_i|$*: intensity of the influence, regardless of its sign;*

— $\omega_i = 0$*: neutral event, with no direct impact.*

Thus, ω_i encodes both the nature and the magnitude of the effect of each event on the health state.

6.5.3 L^1 normalization and properties

Definition 6.4 (Normalized weights)**.** [Original contribution of the author.]

$$\bar{\omega}_i = \frac{\omega_i}{\sum_{j=1}^{N(t)} |\omega_j|} \qquad \textit{so that} \qquad \sum_{i=1}^{N(t)} |\bar{\omega}_i| = 1. \tag{16}$$

Proposition 6.2 (Properties of the L^1 normalization)**.** [Original contribution of the author.]
The normalization (16) gives the operator ζ the following properties:

(i) Scale invariance: for every $c > 0$, replacing ω_i by $c\omega_i$ for all i does not change ζ. Only the relative structure of the contributions is taken into account.

(ii) Stability under the addition of a neutral event: introducing an event of zero weight does not change the value of $S(t)$.

(iii) Control bound: $|S(t)| \leq \max_i |\widetilde{\mathcal{G}}_i(t)|$, which guarantees that the normalization does not artificially amplify the individual contributions.

6.6 Construction of a weight sheet by variational minimization

6.6.1 Introduction

The objective is to predict the weight ω of a new event from its position in the space of attributes. To this end, we construct a continuous surface (the weight sheet) interpolating the observations and satisfying a geometric regularity hypothesis.

6.6.2 Regularity hypothesis

Definition 6.5 (Sheet regularity)**.** [Original contribution of the author.]
Two events with close profiles in $\mathbb{R}^4$ must have close weights. It would be incoherent for an event to have a weight close to -14 while all its neighbors have weights close to 2.

We therefore seek the "flattest" possible surface, compatible with the observed data.

6.6.3 Local roughness and measure of variations

To simplify the exposition and allow a three-dimensional illustration, we place ourselves in the case of two variables (x, y), representing two of the four primary attributes.

Definition 6.6 (Local roughness)**.** *The local roughness of f at (x, y) is:*

$$c(f)(x, y) = \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2. \tag{17}$$

It is zero if and only if f is locally constant. A perfectly flat surface satisfies $c(f) \equiv 0$.

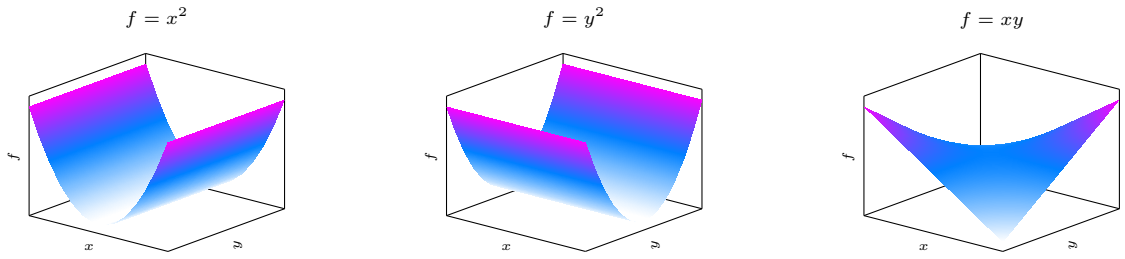
Definition 6.7 (Roughness regimes)**.** *Depending on the direction in which f varies, we distinguish:*

— *Roughness along x only: $\partial f/\partial x$ does not depend on y, but $\partial f/\partial y$ varies. Typical form: $f(x, y) = H(x) + G(y)$ with G non-constant.*

— *Roughness along y only: roles of x and y exchanged.*

— *Roughness along x and y: variations of the partial derivatives in both directions (general case, typical form: $f(x, y) = xy$).*

The following three figures illustrate each of these regimes.



(a) Roughness along x: $\partial f/\partial y = 0$.

(b) Roughness along y: $\partial f/\partial x = 0$.

(c) Cross roughness: variations in both directions.

FIGURE 8 – Three local roughness regimes.

— **(a)** $f = x^2$: roughness along x only, surface constant along y.

— **(b)** $f = y^2$: symmetric case, surface constant along x.

— **(c)** $f = xy$: cross roughness (saddle surface), general case where the partial derivatives vary in all directions.

6.6.4 Usefulness of the local roughness $c(f)$

The visual inspection of a surface f is not always sufficient to identify its zones of strong variation.

The local roughness $c(f)$ acts here as an *instability indicator*: it immediately highlights the regions where the function varies rapidly, even when these variations remain invisible to the naked eye.

6.6.5 Illustration of the local roughness on two representative cases

To illustrate the role of local roughness as an instability indicator, we consider two functions defined on the domain $x + y > 0$:

$$f_1(x, y) = \frac{1}{x+y} \qquad \text{and} \qquad f_2(x, y) = \frac{1}{x+y} + x.$$

The first case highlights a pure singularity, while the second introduces a global linear perturbation. The objective is to observe the robustness of the local roughness with respect to these two types of behavior.

Case 1 — Singular function.

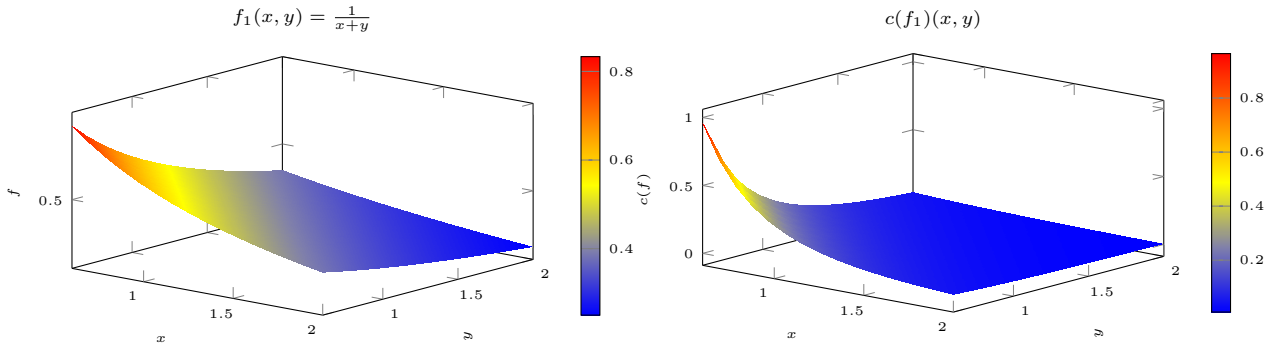


FIGURE 9 – Surface f_1 and associated local roughness.

The function f_1 presents an apparent regularity over a large part of the domain. However, the local roughness

$$c(f_1)(x, y) = \frac{2}{(x+y)^4}$$

reveals an instability strongly concentrated when $x+y \to 0^+$. We thus observe that the critical variations are not directly visible on the surface itself.

Case 2 — Perturbed function.

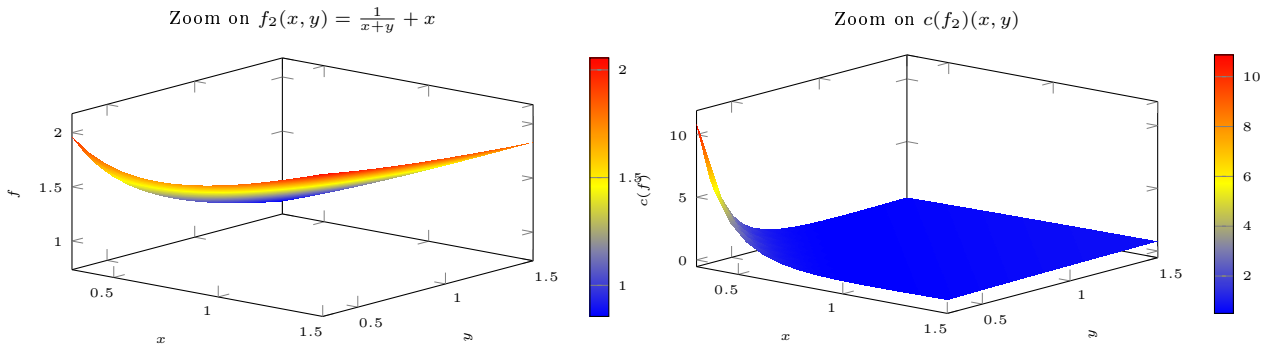


FIGURE 10 – Local zoom on f_2 and its local roughness $c(f_2)$.

The addition of the term x induces a global trend without removing the main singularity. The local roughness becomes:

$$c(f_2)(x,y) = \left(-\frac{1}{(x+y)^2} + 1\right)^2 + \left(-\frac{1}{(x+y)^2}\right)^2.$$

We then observe two distinct contributions:

— a global drift linked to the term x;

— a persistent singularity identical to case 1;

— a clear separation between global structure and local instabilities.

Conclusion. The local roughness $c(f)$ is robust to globally smooth transformations and systematically highlights the zones of structural instability.

6.6.6 Energy functional and optimal sheet

Definition 6.8 (Variation energy)**.** [Original contribution of the author.]
The energy functional of f on a domain $\Omega \subset \mathbb{R}^2$ is:

$$I(f) = \iint_\Omega \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2\right] dx\, dy. \tag{18}$$

The optimal sheet is $f^* = \arg\min I(f)$ under the constraint of interpolating the observed weights $\{(x_j, \omega_j)\}_{j \in C_k}$.

Remark 6.2 (Numerical evaluation of $I(f)$)**.** *In practice, the integral (18) is evaluated by numerical quadrature on the discrete grid τ_d introduced in Section 2.2, from the positions $\{x_j\}$ of the observed events in the cluster C_k (for example by finite differences on these points), rather than computed analytically.*

Proposition 6.3 (Uniaxial roughness with affine component)**.** *We consider a function of the separated form:*

$$f(x,y) = H(x) + G(y),$$

where one of the two components is affine. Without loss of generality, we assume that:

$$H(x) = ax + b, \qquad a, b \in \mathbb{R},$$

while G is a sufficiently regular non-linear function. Since $H'(x) = a$ is independent of b (the variation energy is invariant under vertical translation), the regularity functional decomposes as:

$$I(f) = L_y \int_0^{L_x} a^2\, dx + L_x \int_0^{L_y} [G'(y)]^2 dy = L_y L_x a^2 + L_x \int_0^{L_y} [G'(y)]^2 dy \tag{19}$$

Proof. $\partial f/\partial x = H'(x)$ and $\partial f/\partial y = G'(y)$. Integration over $[0, L_x] \times [0, L_y]$ directly gives (19). □

6.6.7 Normalization by domain size

The energy $I(f)$ grows mechanically with the dimensions of the domain. To compare clusters of different sizes, we normalize by the diameters.

Definition 6.9 (Normalized functional)**.** [Original contribution of the author.]

$$\bar{I}(f) = \frac{1}{L_x L_y} I(f), \tag{20}$$

where L_x (resp. L_y) is the largest distance between two points of the cluster along the x axis (resp. y axis).

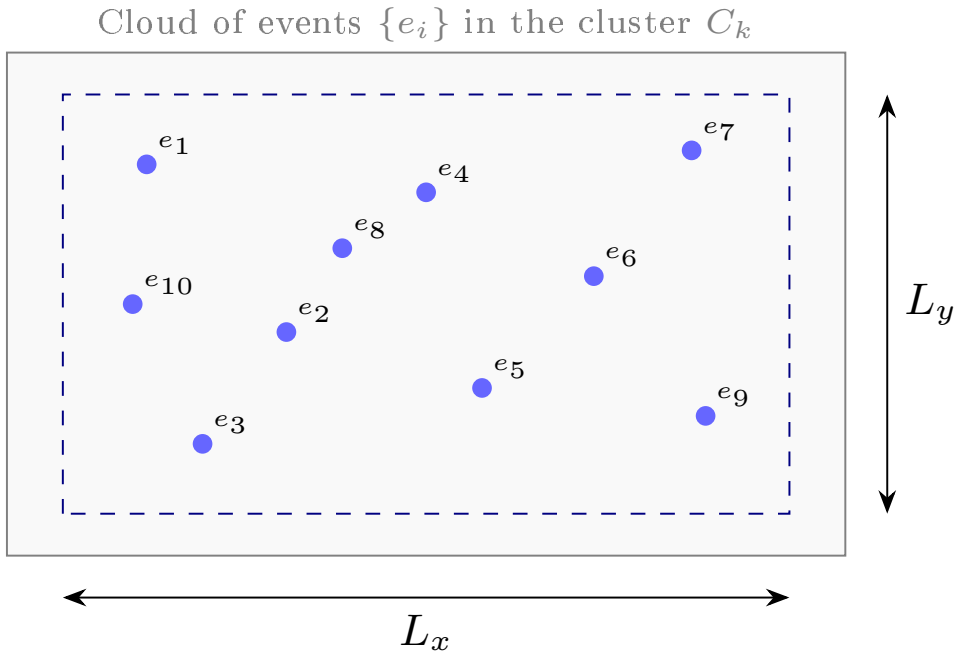


FIGURE 11 – Cloud of events e_i organized in a spatial domain C_k, delimited by a bounding box characterized by L_x and L_y.

Remark 6.3. *In the separable case $f = H(x) + G(y)$, the normalization gives:*

$$\bar{I}(f) = \frac{1}{L_x} \int_0^{L_x} [H'(x)]^2 dx + \frac{1}{L_y} \int_0^{L_y} [G'(y)]^2 dy,$$

that is, the deformation energy per unit length along each axis — a quantity intrinsic to the cluster, independent of its volume.

6.6.8 Default parametric form and prediction

Once the variational problem is posed, the practical question arises of the form of f to optimize. Searching for f^* without shape constraint in the set of all differentiable functions would amount to looking for a needle in a haystack: the space of candidates is too vast to be explored directly.

We therefore propose a default parametric family, flexible enough to adapt to the data while remaining regular. The retained form is an affine combination of the attributes augmented with an interaction term:

Definition 6.10 (Polynomial parametric form of the sheet)**.** [Original contribution of the author.]
On the cluster C_k projected onto two attributes (x, y), we consider the family of polynomials of total degree n:

$$f_{\mathbf{a}}^{(n)}(x, y) = \sum_{\substack{i,j \geq 0 \\ i+j \leq n}} a_{ij}\, x^i y^j, \tag{21}$$

where the coefficients $\mathbf{a} = \{a_{ij}\}$ *are calibrated by minimizing* $\bar{I}(f_{\mathbf{a}}^{(n)})$ *under the constraint of interpolating the observed weights. The number of parameters is* $\binom{n+2}{2} = \frac{(n+1)(n+2)}{2}$*, growing quadratically with* n*. (These coefficients are denoted* a_{ij}*, and not* λ_{ij}*, precisely so as not to conflict with the dilation parameter* λ *introduced later, in Section 7.2.)*

Remark 6.4 (Hierarchy of degrees)**.** *The first levels of the family (21) are:*

- $n = 1$*: affine plane* $a_{00} + a_{10}x + a_{01}y$ *(3 parameters) — no interaction between attributes;*
- $n = 2$*: full quadratic (6 parameters) — covers the three roughness regimes;*
- $n = 3$*: cubic (10 parameters) — captures local curvatures;*
- $n \to \infty$*: convergence to the optimal sheet* f^* *in the variational sense.*

The interpolation accuracy improves with n*, at the cost of a growing risk of overfitting for small clusters.*

Proposition 6.4 (Convergence of the accuracy with n)**.** [Original contribution of the author.]
Let RMSE(n) *be the mean squared prediction error of the weights on a validation set, associated with* $f_{\hat{\mathbf{a}}}^{(n)}$ *calibrated by least squares. Under the hypothesis that the true weights come from a regular function on* Ω*,* RMSE(n) *is decreasing in* n*. The optimal degree* n^* *is defined by:*

$$n^* = \arg\min_n \left[\mathrm{RMSE}(n) + \mu \frac{\binom{n+2}{2}}{m} \right], \tag{22}$$

where $m = |C_k|$ *is the size of the cluster and* $\mu > 0$ *a regularization factor penalizing the relative complexity of the model.*

Remark 6.5 (Intuitive reading of criterion (22))**.** *Intuitively, this criterion expresses a classical bias–variance trade-off: increasing* n *always reduces the fitting error on the points already observed (the sheet fits the training data better and better), but each additional parameter consumes a share of the information available in a cluster of finite size* m*. The term* $\mu\binom{n+2}{2}/m$ *therefore penalizes high degrees in proportion to the number of parameters per available data point: for a small cluster, even a slight RMSE gain is not enough to justify a high degree, whereas a large cluster can afford more parameters without risk of overfitting. Concretely,* n^* *is the smallest degree beyond which the accuracy gain no longer compensates, in practice, the cost — in parameters — of this accuracy. Table 2 (Section 6.9.2) gives a direct numerical illustration: the training RMSE keeps decreasing with* n*, while the validation RMSE stagnates, or even worsens.*

6.7 Asymmetry criterion

Definition 6.11 (Prediction errors)**.** *We distinguish two types of errors in the estimation of interactions:*

- *Underestimation (type I):* $\hat{\mathcal{G}}_i < \mathcal{G}_i^{\mathrm{db}}$*, corresponding to an undervaluation of the real intensity of the event;*
- *Overestimation (type II):* $\hat{\mathcal{G}}_i > \mathcal{G}_i^{\mathrm{db}}$*, corresponding to an overvaluation of this same intensity.*

Underestimation is generally more critical than overestimation, because it can lead to irreversible losses, whereas overestimation mainly induces an opportunity cost. This asymmetry justifies the introduction of a non-symmetric loss function, defined by:

$$\ell(\hat{g}, g) = \begin{cases} c_+ \, (\hat{g} - g)^2 & \text{if } \hat{g} \geq g, \\ c_- \, (\hat{g} - g)^2 & \text{if } \hat{g} < g, \end{cases} \tag{23}$$

where the coefficients c_- and c_+ weight respectively the severity of underestimation and overestimation errors, with $c_- > c_+$ in the case of events with strong negative impact.

The definition of these weights is left to the discretion of the user, in order to adapt to the specificities of the application context.

Remark 6.6 (Practical setting of c_- and c_+)**.** *In the absence of a choice motivated by the context, a reasonable default value is $c_+ = 1$ and $c_- \in [1.5, 3]$: the ratio c_-/c_+ then plays the role of a risk-aversion coefficient, all the higher as one wishes to penalize strongly any underestimation of gravity. This ratio can also be calibrated empirically, by confronting the loss function (23) with a panel of historical events whose real cost of an underestimation is known (observed financial losses, recorded damage), and by choosing c_-/c_+ so as to minimize an operational cost rather than a simple quadratic error. In the absence of such a panel, the choice remains, as indicated, at the discretion of the user — on the same footing as the frame C or the initial weights ω_i (Section 6.5).*

Remark 6.7 (Articulation with per-cluster calibration)**.** *The per-cluster calibration procedure described in Definition 3.7 minimizes, in its current formulation, the symmetric MAPE (5).*

The asymmetry criterion $\ell(\hat{g}, g)$ introduced above constitutes an alternative objective function, particularly relevant for clusters associated with events with strong negative impact (for example $\mathcal{C}_2$ or $\mathcal{C}_3$): substituting ℓ for the MAPE in Equation (6) would make it possible to penalize more severely the underestimations (or conversely, the overestimations) of gravity. This integration is discussed as a perspective in Section 9.

6.8 Complete expression of ζ

$$\zeta(\mathcal{E}(t)) = \tanh\left(\sum_{i=1}^{N(t)} \bar{\omega}_i \, \widetilde{\mathcal{G}}_i(t)\right), \tag{24}$$

where $\bar{\omega}_i$ are the normalized "static" weights (16) and $\widetilde{\mathcal{G}}_i(t)$ integrates the temporal residual gravity (4).

6.9 Preliminary results

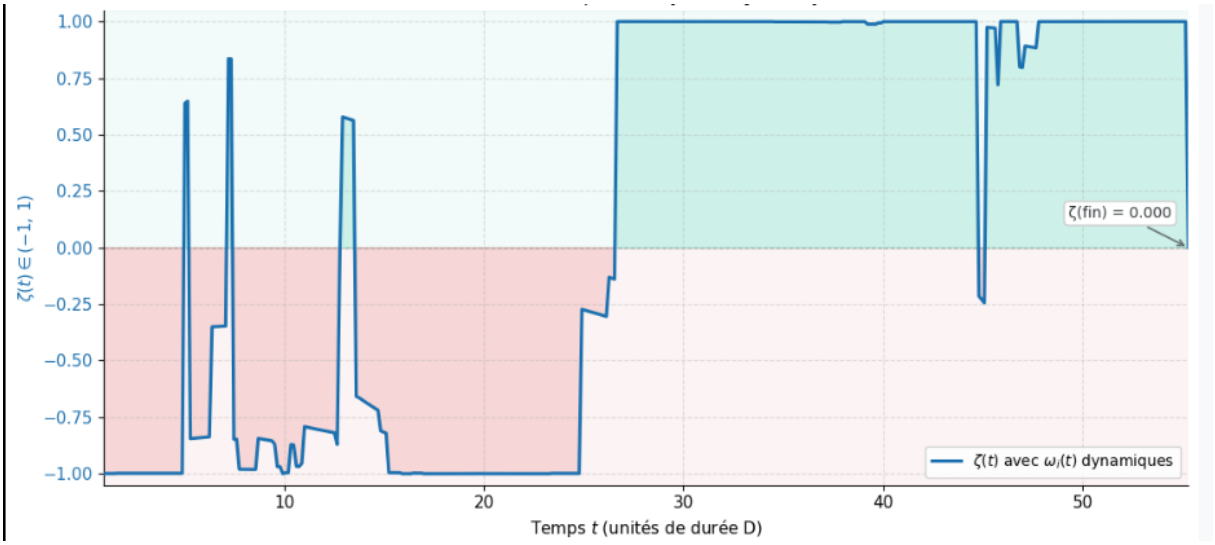


FIGURE 12 – $\zeta(t)$ without dilation ($\lambda = 1$): excessive stagnation at ± 1, motivating the introduction of λ (Section 7.2).

We can now provide a first analysis of the health state of the synthetic environment randomly generated for this preliminary experiment.

The most striking element is the clear demarcation between the left part and the right part of the graph, located around 28 time units.

Moreover, these two regions appear durably frozen at the limit values, which seems unrealistic: it would amount to assuming an environment favorable to the user at the level of 100 % over a prolonged period.

This observation calls for a reduction of the effective intensity of the events. To this end, we introduce in Section 7.2 a dilation factor making it possible to compress the scale of the variations; the results of its integration are presented there and highlight the improvement brought.

However, we can analyze whether the evolution of the health state is consistent with the number of events over time, whether positive or negative. Thus, we expect to observe a majority of negative events in the left part of the time axis, and a majority of positive events in the right part.

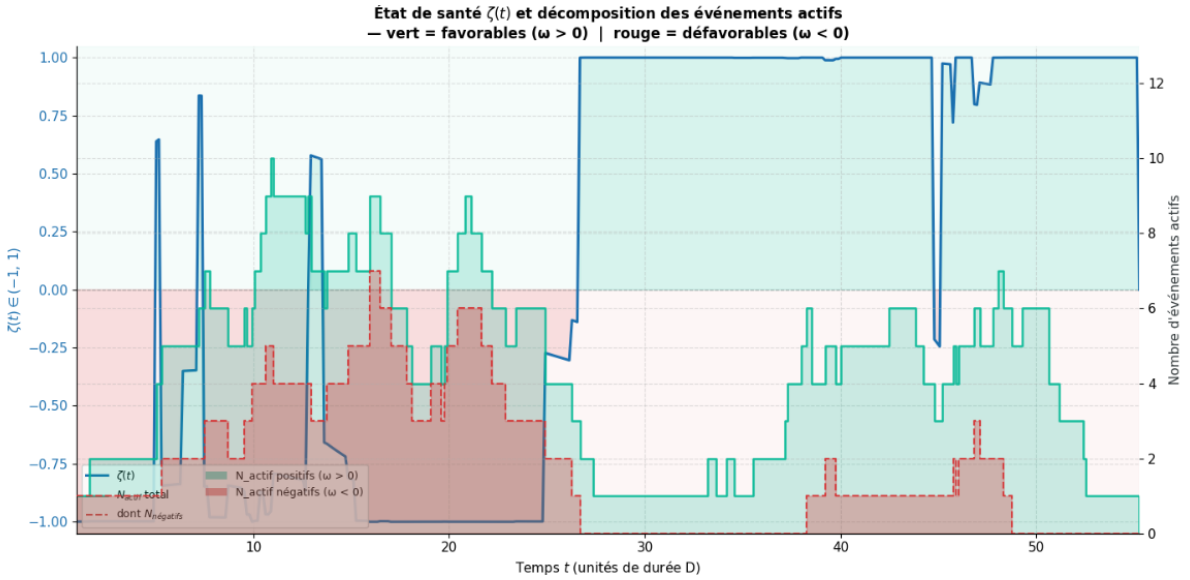


FIGURE 13 – Joint evolution of $\zeta(t)$ and of the number of active events. The qualitative consistency is confirmed.

In accordance with this expectation, a majority of events judged positive for the user is observed in the right part, with however some local variations due to the addition of certain negative events.

Similarly, in the left part, a superiority of negative events over positive events is observed over most of the interval, which confirms the qualitative consistency of the model with the structure of the generated event base.

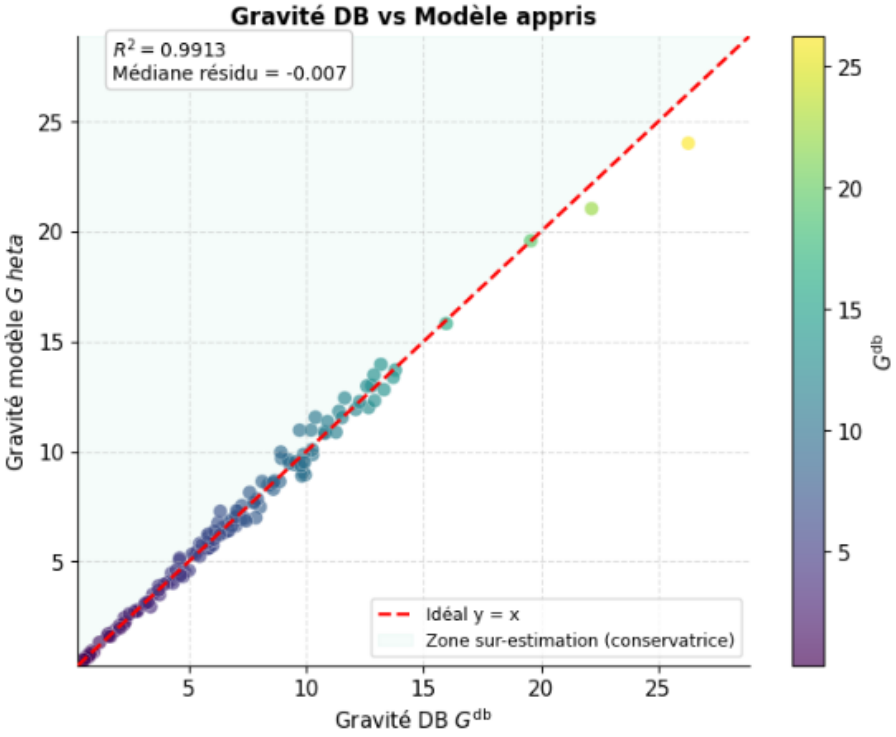


FIGURE 14 – $\hat{\mathcal{G}}$ vs $\mathcal{G}^{\mathrm{db}}$: alignment on $y = x$ with $R^2 = 0.9913$.

The following table summarizes, for each of the three synthetic clusters, the hyperparameters $(\hat{\alpha}_k, \hat{\beta}_k, \hat{\delta}_k)$

estimated by the calibration procedure of Definition 3.7: their detailed reading is the subject of subsection 6.9.1 which follows immediately.

6.9.1 Analysis of the hyperparameters of the gravity law

Cluster	$\lvert C_k\rvert$	$\hat{\alpha}_k$	$\hat{\beta}_k$	$\hat{\delta}_k$
$\mathcal{C}_1$ (Financial)	29	0.7834	0.5608	0.5347
$\mathcal{C}_2$ (Climate)	41	0.7883	0.6098	0.4990
$\mathcal{C}_3$ (Health)	35	0.8157	0.6052	0.4936

We consider the dynamic gravity law defined by:

$$\mathcal{G}(t) = \Gamma\big(\mathcal{D}_{\text{eff}}(t), \mathcal{F}, \mathcal{R}, \mathcal{P}\big) = \frac{\big[\max(\mathcal{D} - (t - t_0), 0)\big]^{\alpha} \cdot \mathcal{F}^{\beta} \cdot \mathcal{R}^{\delta}}{1 + \mathcal{P}^{\gamma}}. \tag{25}$$

The parameters $(\alpha_k, \beta_k, \delta_k)$ estimated per cluster control respectively:

1. the effective temporal decay;
2. the amplification by frequency;
3. the spatial or semantic diffusion.

1. Gaps on the temporal amplitude (α_k).
We observe:

$$\hat{\alpha}_1 = 0.7834, \qquad \hat{\alpha}_2 = 0.7883, \qquad \hat{\alpha}_3 = 0.8157.$$

The maximal relative gap is:

$$\Delta\alpha_{\max} = \frac{0.8157 - 0.7834}{0.7834} \approx 4.1\%.$$

This indicates a strong homogeneity of the temporal decay across clusters, with a slight amplification for the health cluster.

2. Gaps on the sensitivity to frequency (β_k).

$$\hat{\beta}_1 = 0.5608, \qquad \hat{\beta}_2 = 0.6098, \qquad \hat{\beta}_3 = 0.6052.$$

The maximal relative gap is:

$$\Delta\beta_{\max} = \frac{0.6098 - 0.5608}{0.5608} \approx 8.7\%.$$

This result is significant: frequency plays a noticeably larger role in the climate and health clusters than in the financial cluster, which reflects a more marked accumulation dynamic.

3. Gaps on the spatial diffusion (δ_k).

$$\hat{\delta}_1 = 0.5347, \qquad \hat{\delta}_2 = 0.4990, \qquad \hat{\delta}_3 = 0.4936.$$

The maximal relative gap is:

$$\Delta\delta_{\max} = \frac{0.5347 - 0.4936}{0.4936} \approx 8.3\%.$$

We observe here a more marked structural difference: the financial cluster exhibits a wider diffusion, whereas the climate and health clusters are more localized.

Synthesis.
Overall, the gaps between hyperparameters remain moderate (between 4% and 9%), which indicates:

— an overall stability of the gravity law across clusters;

— a specialization mainly carried by β and δ;

— a robustness of the model despite the segmentation.

Thus, the gravity law keeps a coherent structure while capturing dynamic regimes differentiated according to the clusters.

6.9.2 Variational sheets of the static weights ω_i

Beyond the hyperparameters $(\alpha_k, \beta_k, \delta_k, \gamma_k)$ of the gravity law, we now seek to determine whether the static weights ω_i themselves admit an exploitable functional dependence on the descriptive variables of each event, within each of the three clusters.

In full generality, ω_i could depend on the four variables $(\mathcal{D}, \mathcal{F}, \mathcal{R}, \mathcal{P})$ associated with an event. However, since a direct visualization of a four-variable surface is not possible, we restrict the study here to a sheet $f_{\mathbf{a}}^{(n)}(\mathcal{D}, \mathcal{F})$ defined on the two variables $(\mathcal{D}, \mathcal{F})$ only — a projection chosen for its readability in three dimensions, and not because $\mathcal{R}$ and $\mathcal{P}$ would be assumed a priori not relevant. This sheet, a polynomial of total degree n, is calibrated by least squares for each cluster:

$$f_{\mathbf{a}}^{(n)}(\mathcal{D}, \mathcal{F}) = \sum_{i+j \le n} a_{ij}\, \mathcal{D}^i \mathcal{F}^j, \qquad p_n = \binom{n+2}{2} = \frac{(n+1)(n+2)}{2} \text{ parameters.} \tag{26}$$

Remark 6.8 (Reading of formula (26))**.** *This is nothing other than the sheet $f_{\mathbf{a}}^{(n)}$ of Definition 6.10 (Equation (21)), applied to the pair of variables $(x, y) = (\mathcal{D}, \mathcal{F})$. For example, at degree $n = 1$, this simply gives the plane $f_{\mathbf{a}}^{(1)}(\mathcal{D}, \mathcal{F}) = a_{00} + a_{10}\mathcal{D} + a_{01}\mathcal{F}$: the predicted weight then varies in a purely linear way with the duration and the frequency of the event, without interaction term $\mathcal{D}\mathcal{F}$ (the latter appears only from $n = 2$).*

The optimal degree n^* is selected by cross-validation (5 repeated 80/20 draws), by minimizing a penalized criterion $\mathrm{RMSE}(n) + \mu\, p_n/m$ ($\mu = 0.05$) in order to avoid over-parametrization when the number of points m available per cluster is small (see Remark 6.5 for an intuitive reading of this criterion).

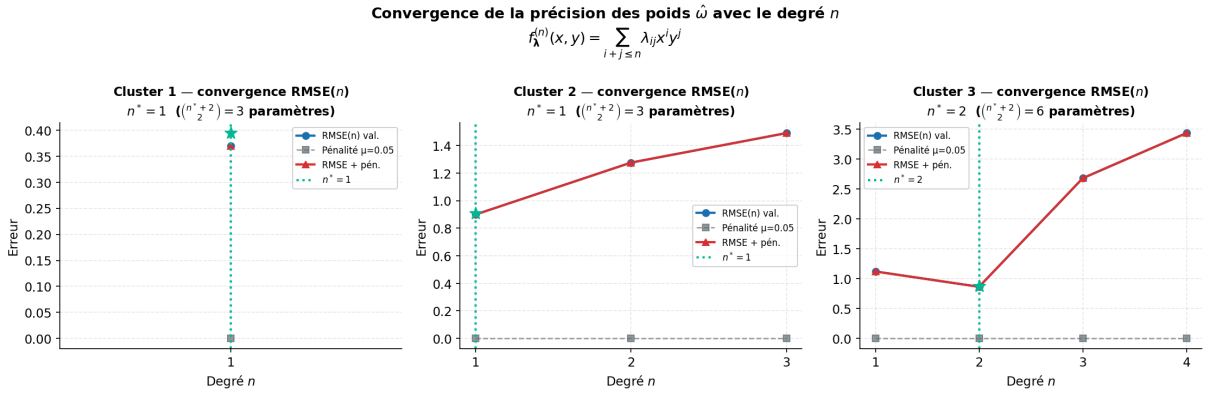


FIGURE 15 – Convergence of the cross-validation RMSE as a function of the degree n, per cluster.

Cluster	m (test)	n^*	p_{n^*}	RMSE(n^*)	$\bar{I}(f^*)$	$\hat{\omega}(\mathcal{D}{=}4, \mathcal{F}{=}2)$
$\mathcal{C}_1$ (Financial)	6	1	3	0.3695	0.3514	−0.3711
$\mathcal{C}_2$ (Climate)	17	1	3	0.8990	0.0029	+0.3981
$\mathcal{C}_3$ (Health)	22	2	6	0.8632	1.6994	−0.0970

TABLE 1 – Optimal degree, cross-validation error, normalized energy $\bar{I}(f^*)$ and weight prediction for a hypothetical event $(\mathcal{D}, \mathcal{F}) = (4, 2)$, per cluster.

The weights ω_i observed over the complete set cover the interval $[-1.49, 1.48]$, with a near-perfect balance between positive and negative weights (74 positive / 76 negative out of 150 events). The validation RMSE obtained (between 0.37 and 0.90) must be compared with the theoretical standard deviation of a uniform law over this interval, $\sigma = 3/\sqrt{12} \approx 0.87$: for the Climate and Health clusters, the validation RMSE is of the same order as σ, which indicates that **the retained $(\mathcal{D}, \mathcal{F})$ projection does little better than a constant model** (the local mean of the cluster).

This observation concerns specifically this two-variable projection: it does not, by itself, exclude a dependence of ω_i on $\mathcal{R}$ and/or $\mathcal{P}$, which the 3D representation does not allow us to explore directly. The predictions for a new event $(\mathcal{D}, \mathcal{F}) = (4, 2)$ differ strongly from one cluster to another in sign and

amplitude (-0.37, $+0.40$, -0.10), which is consistent with quasi-flat sheets calibrated on few points: the prediction mainly reflects the local mean of the cluster rather than a true geometric dependence on $(\mathcal{D}, \mathcal{F})$.

This numerical reading can be completed by a direct visual overview of the shape of the sheets $f^*(\mathcal{D}, \mathcal{F})$ calibrated at the optimal degree n^* for each of the three clusters (Figure 16). The observed points are superimposed on the sheet, colored green when $\omega_i \geq 0$ (favorable event) and red when $\omega_i < 0$ (unfavorable event).

We find there, consistently with the numerical results of Table 1, little-rugged surfaces: an almost horizontal plane for the Financial and Climate clusters ($n^* = 1$), and a slight curvature for the Health cluster ($n^* = 2$), with no marked relief. The positive and negative points appear intermingled above and below the sheet rather than separated by a clear boundary along $\mathcal{D}$ or $\mathcal{F}$ — a visual signature of the absence of exploitable geometric structure noted above.

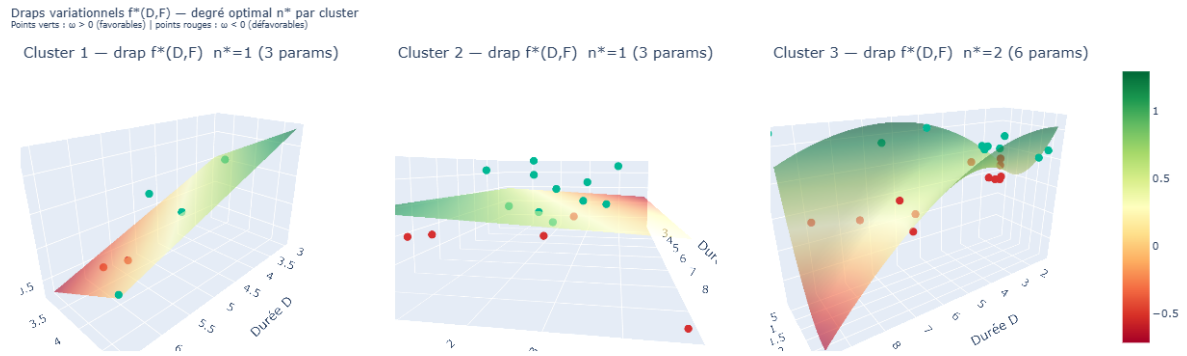


FIGURE 16 – Variational sheets $f^*(\mathcal{D}, \mathcal{F})$ at the optimal degree n^*, per cluster (Financial, Climate, Health).

An apparent paradox: accuracy does not improve with n. One might expect a higher-degree polynomial, with more parameters, to always approximate the data better. This is indeed true for training error, but not for validation:

Cluster	n	RMSE train	RMSE validation
$\mathcal{C}_2$ (Climate)	1	0.6054	0.8530
$\mathcal{C}_3$ (Health)	1	0.8150	0.9748
$\mathcal{C}_3$ (Health)	2	0.5777	0.9523

TABLE 2 – Train/validation diagnosis: the training RMSE decreases with n, but the validation RMSE stagnates and the gap between the two doubles between $n = 1$ and $n = 2$ for the Health cluster.

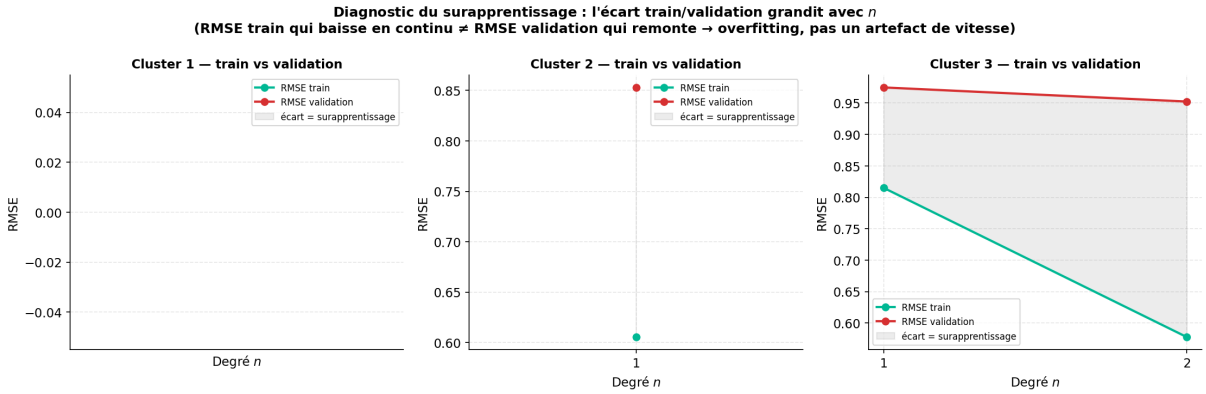


FIGURE 17 – Train RMSE (green) vs validation RMSE (red) per cluster; the grey gap, a signature of overfitting, grows with n.

The training RMSE of the Health cluster decreases from 0.815 to 0.578 between $n = 1$ and $n = 2$, confirming that the polynomial fits the *seen* points better and better. But the validation RMSE does not follow this trend (marginal improvement from 0.975 to 0.952), while the train/validation gap doubles ($0.160 \rightarrow 0.375$): this is the classical signature of **overfitting** — the additional degree brings no new signal, it absorbs the sampling noise specific to the training points.

The methodological reservation formulated above on the restriction to $(\mathcal{D}, \mathcal{F})$ can in fact be lifted in the precise case of this preliminary experiment: the very construction of the synthetic base shows that the static weights are generated there by $\omega_i \sim \mathcal{U}(-1.5, 1.5)$, **independently of the four variables $\mathcal{D}$, $\mathcal{F}$, $\mathcal{R}$ and $\mathcal{P}$.**

There therefore exists, by construction, no underlying function $f(\mathcal{D}, \mathcal{F}, \mathcal{R}, \mathcal{P})$ to recover — the real relation is pure noise rather than a smooth surface, whichever pair of variables is chosen for the projection. This is therefore not here an artifact of the choice of $(\mathcal{D}, \mathcal{F})$ rather than $(\mathcal{R}, \mathcal{P})$: on real data where this generation information would not be available, the same conclusion should be verified by crossing several projections (or, better, by fitting a sheet on the four variables without going through a direct visualization) before being retained. The penalized criterion moreover correctly selects low degrees ($n^* \in \{1, 2\}$), which constitutes the expected behavior of the safeguard in the face of a target without exploitable structure, rather than a limit of the method.

7 Model extensions

7.1 Introduction

The preliminary results of Section 6 highlighted three practical limitations of the base model: a premature saturation of ζ at the bounds ± 1, an insufficient sensitivity to local variations when the number of active events becomes high, and the absence of an automatic calibration mechanism for the hyper-parameters.

The following sections successively address these three points, before tackling the extension of the model to events corresponding to no pre-existing cluster.

7.2 Sensitivity adjustment by dilation

The first simulations highlighted a phenomenon of premature saturation of ζ at the bounds ± 1, limiting the discrimination capacity in strongly extreme regimes. In order to control this behavior, we introduce a dilation parameter.

Definition 7.1 (Dilation)**.** [Original contribution of the author.]
The dilated state function is defined by:

$$\zeta_\lambda(\mathcal{E}(t)) = \tanh\left(\frac{S(t)}{\lambda}\right), \qquad \lambda > 0. \tag{27}$$

where λ controls the saturation scale of the response.

We will distinguish between the λ of this definition and the $\lambda_{i,j}$ of part 6.8.2.

Remark 7.1 (Notation)**.** *In order to avoid any ambiguity with the coefficients of the variational sheet of Section 6.6, the latter have been denoted a_{ij} (and not λ_{ij}): λ is thus, in the whole remainder of the article, reserved exclusively for the dilation parameter above.*

Proposition 7.1 (Properties of ζ_λ)**.** *The family of functions ζ_λ satisfies the following properties:*

- *Local sensitivity: $\zeta'_\lambda(0) = \dfrac{1}{\lambda}$, which shows that the sensitivity around the origin is inversely proportional to λ;*
- *Saturation threshold: $\zeta_\lambda \approx 0.95$ as soon as $|S| \geq 1.83\,\lambda$, indicating a direct control of the non-linearity zone;*
- *Diffuse regime: when $\lambda \to +\infty$, we have $\zeta_\lambda(\mathcal{E}(t)) \to 0$, which corresponds to a progressive loss of sensitivity and hence of information.*

λ	Derivative at 0	$\lvert S\rvert$ for $\zeta = 0.95$	Interpretation
0.5	2.00	0.91	Very sensitive, fast saturation
1	1.00	1.83	Standard (plateau problem)
5	0.20	9.16	Moderate filter, used in practice
20	0.05	36.6	Strong filter
100	0.01	183	Quasi-linear

TABLE 3 – Effect of λ on the sensitivity of ζ_λ.

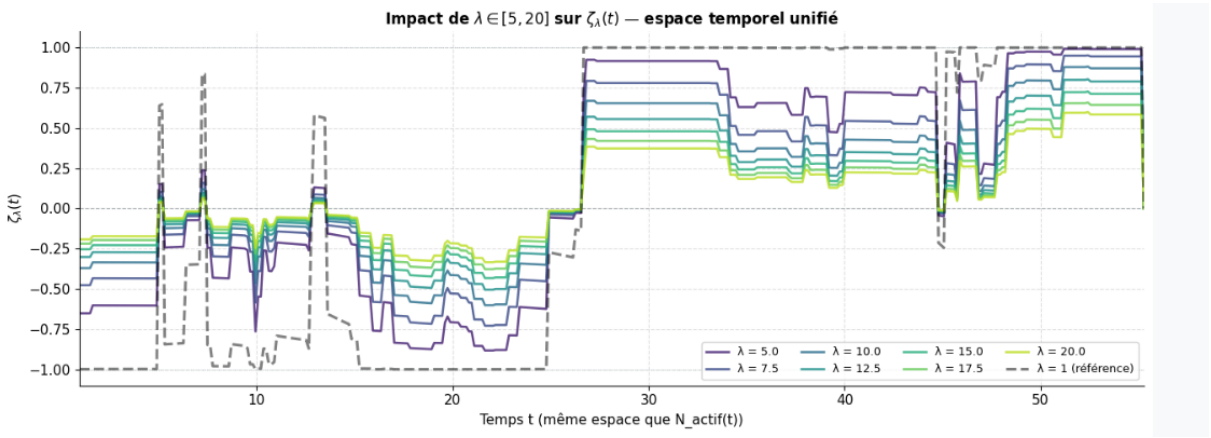


FIGURE 18 – $\zeta_\lambda(t)$ for different values of λ: progressive attenuation of the saturation effects at the extremities.

The analysis of the results shows that the introduction of the parameter λ makes it possible to effectively control the phenomenon of saturation of the health state at the bounds ± 1. Intermediate values such as $\lambda \in \{7.5, 10, 12.5\}$ appear as a relevant compromise, keeping a legible dynamic while avoiding an excessive crushing of the information.

Time-scale problem.

A difficulty appears however when the environment accumulates a large number of events, even of low intensity. In this case, the cumulative contribution tends to grow artificially, which calls for an adaptation of the regularization parameter.

To address this phenomenon, we introduce a dependence of the parameter λ on the temporal structure of the environment:

$$\lambda(t) = f\big(N(t)\big), \tag{28}$$

where $N(t)$ represents the number of active events at instant t.

Choice of the model.

Among the different possible forms for f, we retain a deliberately simple and interpretable version:

$$\lambda(t) = N(t). \tag{29}$$

This choice is justified by several considerations:

— $N(t)$ is a direct measure of the informational density of the system;

— the linear dependence ensures an adaptation proportional to the local complexity;

— the absence of additional parameters limits the risks of overfitting;

— the reading of λ remains directly linked to an observable quantity.

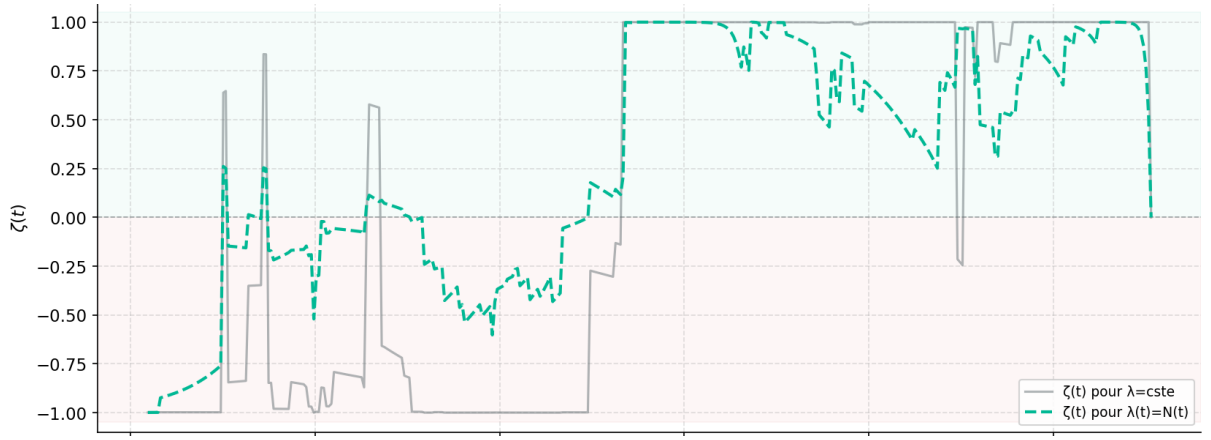


FIGURE 19 – $\zeta_\lambda(t)$ for $\lambda(t) = 1$ and $\lambda(t) = N(t)$

Unlike the configuration with a constant dilation factor, the results obtained here highlight a distinctly more nuanced behavior of the dynamics of $\zeta_\lambda(t)$.

With a fixed λ, the weighted sum $S(t)$ can exceed the saturation threshold of tanh as soon as the number of active events becomes sufficiently high, artificially compressing ζ_λ into a reduced range. The introduction of an adaptive parameter $\lambda(t) = \alpha\, N(t)$ directly compensates for this accumulation: as $N(t)$ grows, the saturation threshold $\lambda(t)$ grows proportionally, restoring a natural dynamic scale and a better distribution of the variations over $[-1, 1]$.

However, local plateaus over short time windows persist. This phenomenon is explained by phases where $N(t)$ varies slowly: the denominator $\lambda(t)$ remains quasi-constant, and series of events of the same polarity can again saturate tanh. To remedy this, we propose to make saturation exponentially difficult to reach by introducing a growing second-order penalization:

Definition 7.2 (Penalized adaptive dilation)**.** [Original contribution of the author.]
We define the adaptive dilation parameter by:

$$\lambda(t) = \frac{\alpha\, N(t)}{1 + \beta\, |\zeta_\lambda(t-1)|}, \qquad \alpha > 0,\ \beta > 0, \tag{30}$$

where $t-1$ denotes the previous instant, $\alpha > 0$ is the global calibration factor, and $\beta > 0$ controls the intensity of the anti-plateau correction.

The recurrence is initialized by the initial condition $\zeta_\lambda(t_0 - 1) := 0$, corresponding to a neutral environment before the first observation.

When $|\zeta_\lambda(t^-)| \to 1$, λ decreases, making tanh steeper and forcing ζ to return toward the center. This mechanism encodes precisely the *principle of impossibility of absolute extremes* introduced in Section 6.4.

— Neutral zone ($|\zeta_\lambda(t^-)| \approx 0$): the denominator is ≈ 1, the correction is negligible and $\lambda(t)$ reduces to $\alpha\, N(t)$;

— Imminent saturation ($|\zeta_\lambda(t^-)| \approx 1$): the denominator reaches $1 + \beta$, reducing $\lambda(t)$ and making any further crossing toward ± 1 progressively difficult.

In other words, the closer ζ_λ gets to the bounds, the more the model "resists" reaching them — which precisely encodes the principle of impossibility of absolute extremes introduced in Section 6.4.

Remark 7.2. The two limiting cases make it possible to recover the previous formulations:

— *$\beta = 0$: $\lambda(t) = \alpha\, N(t)$, dilation proportional to the number of events without penalization;*

— *$\beta = 0$ and $N(t) \equiv N_0$ constant, with $\alpha = \lambda_0 / N_0$: $\lambda(t) = \lambda_0$, we recover the constant factor (Section 7.2).*

Remark 7.3 (Non-crossing condition)**.** *The preceding mechanism slows down the progression toward ± 1, but does not formally prevent it in full generality: ζ_λ can keep progressing toward the bounds as long as $S(t)$ grows faster than $\lambda(t)$. More precisely, if $1 + \beta|\zeta_\lambda(t-1)| > \alpha N(t)$ — that is, when the anti-plateau correction in the denominator outweighs the raw effect of $N(t)$ in the numerator —, then $\lambda(t)$ is smaller than what it would be without penalization, which on the contrary accelerates the local saturation of* tanh *instead of slowing it. The parameter β must therefore be calibrated so that this inequality holds only in the immediate neighborhood of $|\zeta_\lambda| \to 1$ (where the sought effect — slowing the approach to the bound — remains coherent), and not over an extended range of values of ζ_λ, failing which the penalization would locally produce the opposite effect to the one sought. It is precisely this compromise that motivates the choice of (α, β) illustrated below (Figure 21).*

Remark 7.4. *This extension may prove relevant in situations where an excessive presence of recurrent plateaus is observed in the dynamics of $\zeta_\lambda(t)$. The introduction of α then makes it possible to finely adjust the intensity of the dilation and to restore a better sensitivity to local variations.*

Validation of the adaptive dilation factor.
The objective of this experiment is to verify that the introduction of the adaptive dilation factor makes it possible to limit the saturation phenomena of $\zeta_\lambda(t)$ at the extreme bounds -1 and 1.

$$\lambda(t) = \frac{\alpha\, N(t)}{1 + \beta\, |\zeta_\lambda(t-1)|}$$

In order to illustrate the expected behavior of this dilation factor, we propose a simulated graph. This graph, deliberately idealized, is not intended to faithfully reproduce a real case: it serves only to highlight, in a pedagogical manner, the sought effect.

A comparison with a version calibrated on subsequent data will be necessary in order to precisely assess the effectiveness of the mechanism under real conditions.

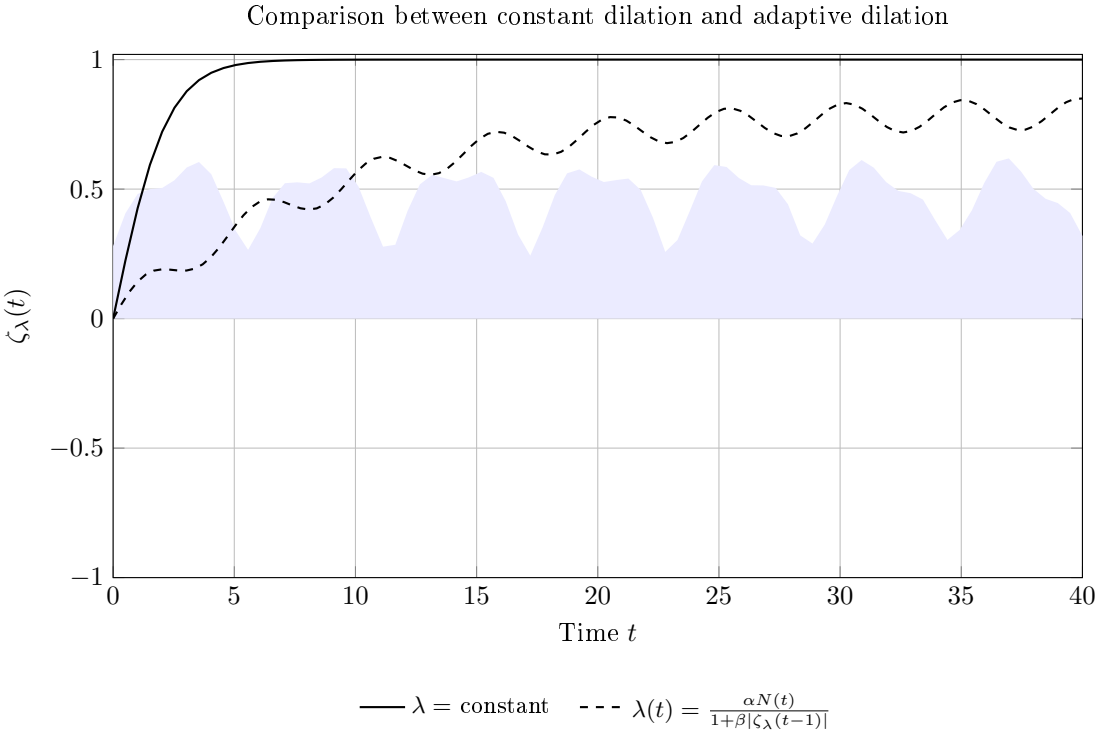


FIGURE 20 – Comparison between a constant dilation and an adaptive dilation.

In order to illustrate this behavior, we compare:

— a model with constant dilation;

— a model with adaptive dilation depending on the number of active events $N(t)$.

In the case of a constant dilation factor, strong accumulations of events quickly lead to plateaus, which strongly reduces the dynamics of the signal and causes a loss of information on the local variations of the environment.

The transparent pastel zone represents here the evolution of the number of active events in the environment. Ideally, the dilation factor thus makes it possible to flatten the intensity peaks in order to avoid any excess or artificial overload in the evaluation of the health state.

This model requires however determining a pair of hyperparameters (α, β). This choice is not trivial: depending on the context and the environment considered, this pair may vary significantly (see Remark 7.3 for the constraint that this pair must respect).

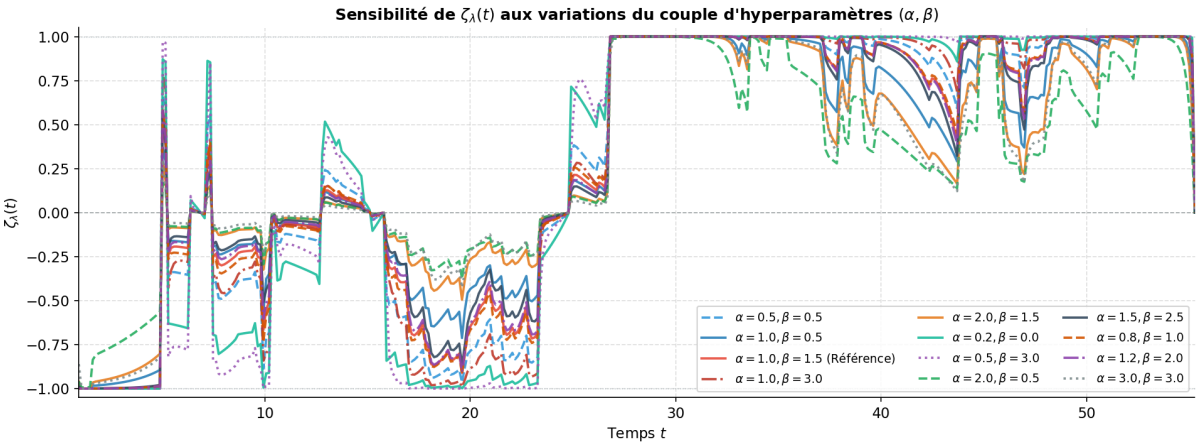


FIGURE 21 – Health state according to the pair (α, β).

In our case, the pair $\alpha = 2$ and $\beta = 0.5$ seems to constitute the best compromise, insofar as it produces the curve most representative of the expected behavior. In other words, this pair is the one that delays the longest the appearance of plateaus.

7.3 Temporal smoothing by interpolation

7.3.1 Lagrange and Runge phenomenon

Definition 7.3 (Lagrange polynomial)**.** [Standard result from the literature.]

$$\hat{\zeta}(t) = \sum_{k=0}^{n} \zeta_k \, \ell_k(t), \qquad \ell_k(t) = \prod_{j \neq k} \frac{t - t_j}{t_k - t_j}. \tag{31}$$

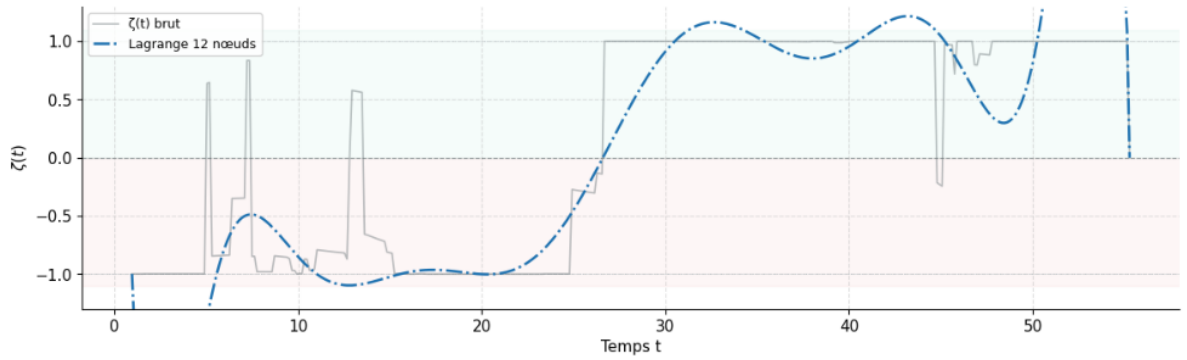


FIGURE 22 – Lagrange smoothing: outputs outside $[-1, 1]$ (Runge phenomenon).

As we can clearly see, Lagrange interpolation leads to a smoothing that largely exceeds the domain of study of the health state, defined on the interval $[-1, 1]$. Thus, we cannot rely on this method in our analysis framework.

This is why we turn to another smoothing method, namely cubic splines.

7.3.2 Cubic splines — recommended method

Definition 7.4 (Natural cubic spline)**.** [Standard result from the literature.]
[5] C^2 function, piecewise polynomial (degree 3), minimizing $\int [\hat{\zeta}'']^2 \, dt$.

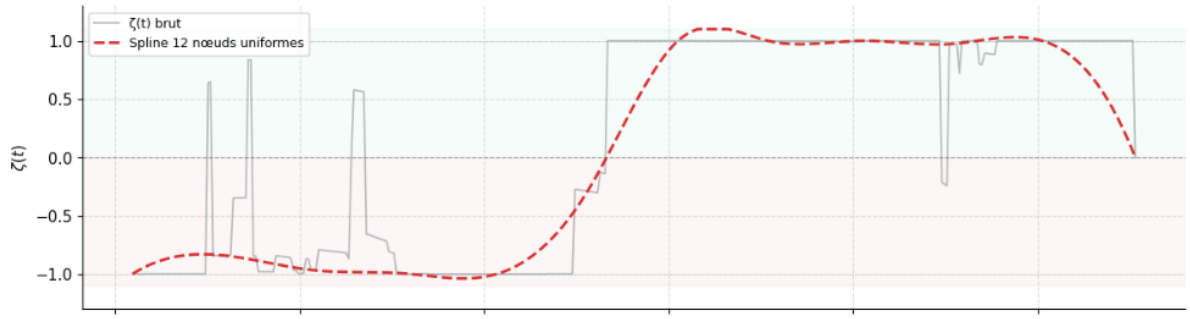


FIGURE 23 – Cubic spline smoothing: curve bounded in $[-1, 1]$.

Unlike Lagrange interpolation, the domain of definition of the health state is respected, despite the appearance of slight relative deviations of the order of a few percent. In a qualitative analysis framework, these deviations remain acceptable insofar as the main objective is to identify the zones where the environment is strongly favorable or unfavorable to the user.

However, an important limitation appears when one is interested in variations over very short time windows. These local fluctuations are not correctly captured by the smoothing induced by cubic splines.

This limit comes directly from the choice of the nodes, which are distributed uniformly over the whole time axis. Such a distribution does not allow the resolution to be adapted to zones presenting rapid variations or local discontinuities.

It therefore becomes necessary to consider an adaptive densification of the nodes in the regions where the dynamics become strongly non-linear. This idea leads naturally to the next section.

7.3.3 Splines adaptive to discontinuities

The idea that follows — densifying the measurement points around an abrupt variation in order to better capture its shape, rather than sampling the whole domain uniformly — is reminiscent of the practice of pH-metric titration in chemistry, where measurements are tightened around the equivalence point in order to better capture the pH jump.

Definition 7.5 (Adaptive nodes)**.** [Original contribution of the author.]

$$T_{\text{disc}} = \{t_k, t_{k+1} \mid \Delta_k > \Delta_{\text{thr}}\}, \qquad \Delta_k = |\zeta(t_{k+1}) - \zeta(t_k)|. \tag{32}$$

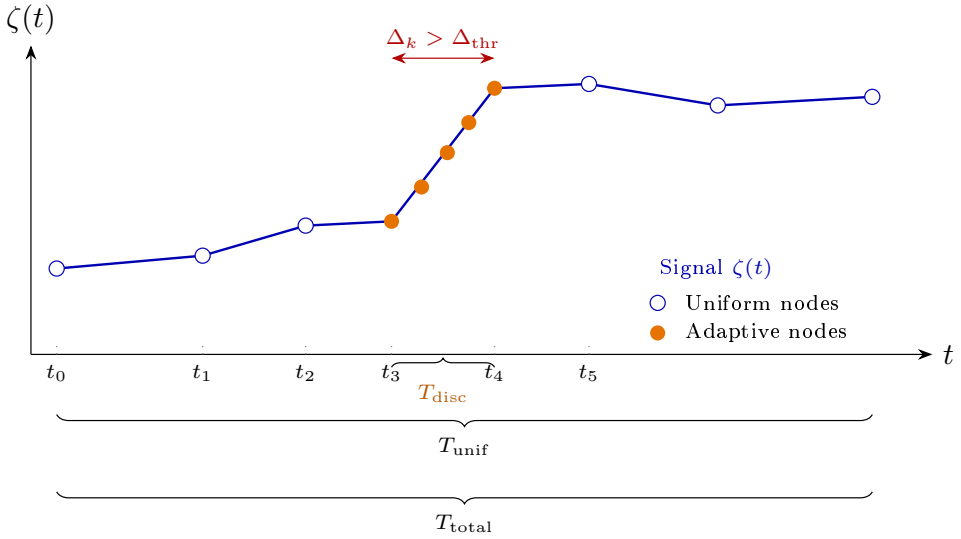


FIGURE 24 – Grid enrichment: adaptive nodes injected around the jumps $\Delta_k > \Delta_{\text{thr}}$.

As the illustration shows, the detection of a discontinuity in the evolution of the health state leads to a local refinement of the time discretization. The idea consists in increasing the density of nodes in the zones where the dynamics present abrupt variations, in order to improve the resolution.

This strategy makes it possible to better capture the local behaviors of the signal and to improve the quality of the smoothing in strongly non-linear regions.

We denote by N_{unif} the number of nodes initially distributed uniformly over the time interval studied. For each detected discontinuity point t_k, we introduce an additional set of nodes, whose cardinality is denoted n_k, in order to locally refine the grid.

The total number of nodes is then written:

$$N_{\text{total}} = N_{\text{unif}} + \sum_{k \in \mathcal{D}} (n_k - 1), \tag{33}$$

where $\mathcal{D}$ denotes the set of indices of the detected discontinuities.

The term $(n_k - 1)$ makes it possible to correct the double counting of the central node already present at t_k, when adding the local refinement points.

Thus, the time discretization becomes adaptive: its density depends directly on the local complexity of the signal $\zeta(t)$.

We will now combine the results of this section with those of the section devoted to the sensitivity adjustment by dilation (Section 7.2).

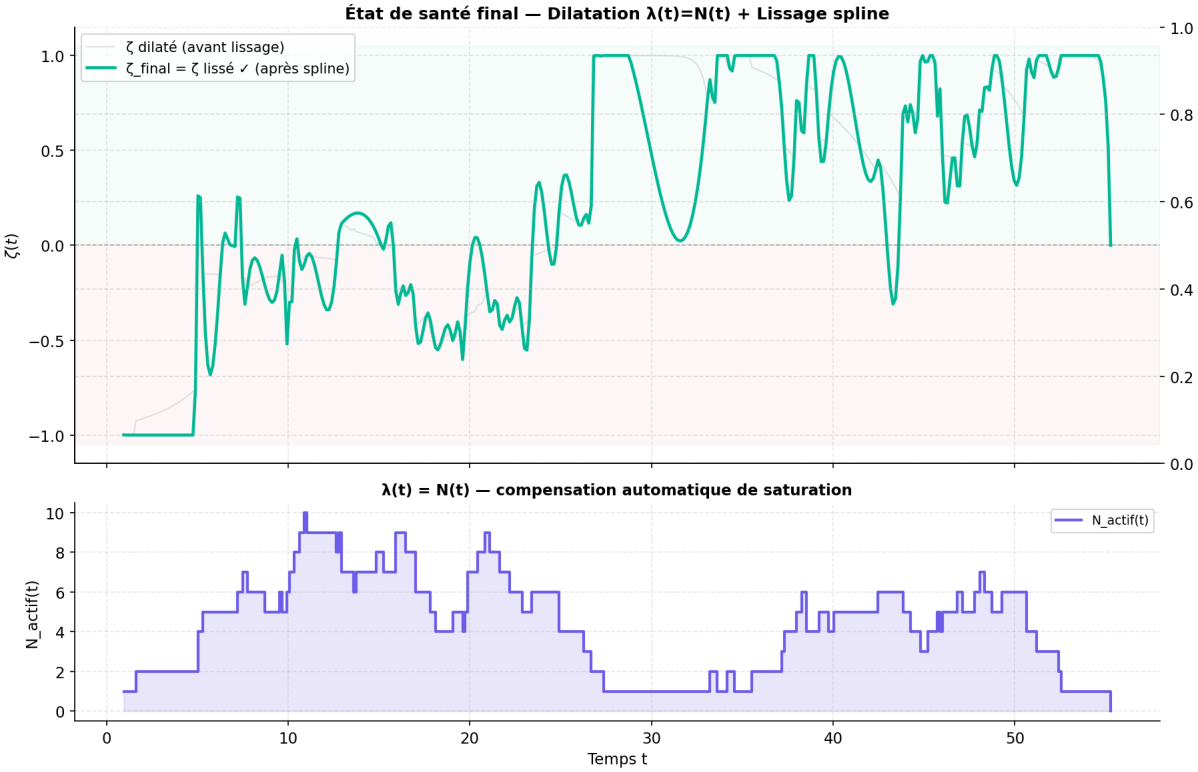


FIGURE 25 – Temporal smoothing of the health state by adaptive cubic spline.

We can conclude that the adaptive smoothing makes it possible to faithfully follow the curve, even if a larger or smaller gap can be observed in places. Indeed, the maximal absolute offset ($|\Delta\zeta|$) culminates at about 1.0 (i.e. 50 % of the total amplitude of the signal) around $t = 31$, where the smoothed curve drops to 0.0 while the original signal saturates at 1.0. Moreover, multiple other major deviations, between 0.3 and 0.7, are observable throughout the signal, notably around $t = 38$ and $t = 44$.

This result is satisfactory: apart from these few gaps, the smoothing fulfills its role without omitting fine information, which was precisely lacking in the previous versions of the smoothing. It also has the advantage of strictly respecting the domain of definition of the health state, the curve always remaining between -1 and 1, including when plateaus form.

The main performance obtained is thus a more faithful approximation of the real curve, while preserving the subtleties of the latter.

Qualitative analysis and critical discussion:
However, from a purely qualitative point of view, this smoothing presents an important compromise that deserves to be underlined. If the ability of the spline to stick to rapid dynamics is positive, it comes at the expense of local stability.

The appearance of deep and unjustified troughs (notably at $t = 31$ and $t = 44$) where the original signal is nevertheless stable or textured shows a phenomenon of parasitic oscillations typical of splines that are poorly constrained on abrupt transitions.

In conclusion, the smoothing is excellent for capturing the reactivity and the extrema of the system, but deficient in restoring the flatness of plateaus, by introducing artificial high-frequency noise where the real signal was smooth.

7.4 Learning by K-fold cross-validation

7.4.1 Introduction

The MAPE (Definition 3.6) is chosen in our framework because it makes it possible to quantify the error in a relative way and therefore independently of the scale of the data. This property is particularly relevant for the evaluation of the operator Γ, whose outputs correspond to a normalized gravity of the health state.

Moreover, its expression as a percentage facilitates the direct interpretation of the model's performance, which allows a clear comparison between different configurations (global or per cluster), as in the K-fold cross-validation defined in Definition 7.6. Let us recall, for the readability of what follows, that $\hat{\theta}_k = (\hat{\alpha}_k, \hat{\beta}_k, \hat{\delta}_k, \hat{\gamma}_k)$ denotes, as in Definition 3.7, the set of parameters calibrated for the cluster C_k.

Definition 7.6 (K-fold cross-validation)**.** [Standard result from the literature.]
(applied here to Γ*):*

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{K} \sum_{k=1}^{K} \mathrm{MAPE}(\Gamma_\theta, F_k). \tag{34}$$

7.4.2 Analysis of the model's performance

This comparison pursues a double objective: to confirm, a posteriori, the relevance of the choice — made from Section 3.4 on and announced in Remark 3.4 — of calibrating Γ per cluster rather than globally, a choice on which all the following sections rely; and to quantify the gain brought by the successive refinements introduced since (dilation, adaptive smoothing, etc.), which all rely on this same structuring into clusters.

Model	MAPE	R^2
Global (no cluster)	4.32%	0.9913
Per cluster	4.10%	—
Best (C_3)	3.35%	—

The results obtained show that the model has a very good prediction capacity, with low relative errors across all the experiments carried out.

1. Global model.
The model trained without separation into clusters reaches a mean absolute relative error:

$$\mathrm{MAPE} = 4.32\%.$$

This value indicates that the mean gap between the predictions and the observed values remains below 5%, which reflects a good overall accuracy of the model.

The associated coefficient of determination is:

$$R^2 = 0.9913.$$

Thus, more than 99% of the observed variance is explained by the model, which attests to a very strong consistency between the predictions and the real data.

2. Effect of structuring by clusters.
The introduction of a separation of the data by clusters slightly improves the performance:

$$\mathrm{MAPE} = 4.10\%.$$

This decrease in the error suggests that the dynamics specific to certain categories of events are better captured when a specialized model is used locally.

In other words, the structuring into clusters makes it possible to reduce the overall heterogeneity of the data and to obtain estimates better adapted to the specific behaviors of each group.

3. Case of cluster C_3.
The cluster C_3 is the best-performing case:

$$\mathrm{MAPE} = 3.35\%.$$

Such a value indicates that the events belonging to this cluster probably present:

— a more homogeneous internal structure;

— more regular relations between the variables;

— or a dynamic that is more easily modelable.

4. Global interpretation.
Overall, the different metrics obtained confirm:

— the numerical stability of the model;

— its ability to faithfully reproduce the observed behaviors;

— the interest of a local approach by clusters;

— as well as the relevance of the regularity hypotheses introduced previously.

The residual gaps observed come mainly from:

— zones presenting strong local variations;

— rapid transitions between different dynamic regimes;

— or atypical events that are hard to interpolate.

7.5 Learning by Q-Learning

The previous sections highlighted several hyperparameters whose manual tuning is delicate: the pair (α, β) of the adaptive dilation (Section 7.2), the threshold Δ_{thr} of the adaptive nodes (Section 7.3.3), or the exponents $(\alpha_k, \beta_k, \delta_k, \gamma_k)$ of the per-cluster gravity law (Section 3.4), for now calibrated by direct minimization of the MAPE (Equation (6)). We explore in this section the possibility of entrusting this tuning to an agent learning by reinforcement rather than to a simple direct optimization procedure, in the hope of automating and refining this calibration.

Q-Learning is a reinforcement learning algorithm, said to be *model-free* ("without a transition model provided a priori"): it requires no prior knowledge of the transition model of the environment, that is, of the function $P(s' \mid s, a)$ giving the probability of moving from state s to state s' under the effect of action a, a quantity which, conversely, classical dynamic programming methods require.

Its objective is to learn an optimal action policy that maximizes the cumulative reward of an agent interacting with its environment, without requiring prior knowledge of the physical or logical laws governing this environment.

7.5.1 Fundamental principles

Q-Learning relies on the evaluation of the quality of a given action in a specific state. The agent maintains and updates an action-state value function, denoted $Q(s, a)$. This function represents the mathematical expectation of future rewards if the agent executes action a from state s, then follows the optimal policy for the following steps.

The functioning of the algorithm is organized around the following key concepts:

— The State ($s \in \mathcal{S}$): The numerical or symbolic representation of the current situation of the agent in its environment.

— The Action ($a \in \mathcal{A}$): The decision made by the agent among a set of possible actions in state s.

— The Reward (r): The scalar signal returned by the environment after the execution of the action, evaluating its immediate efficiency (positive for a success, negative for a penalty).

— The Policy (π): The overall strategy of the agent, defining which action to choose for each state. The optimal policy is denoted π^*.

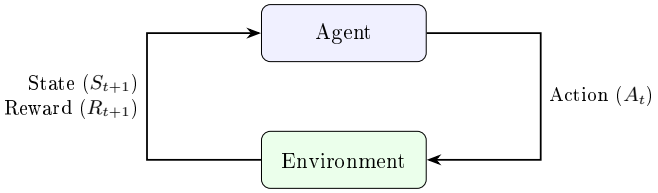


FIGURE 26 – Classical diagram of the interaction loop between the agent and the environment in reinforcement learning.

7.5.2 The Bellman equation and the update of values

In discrete environments, the Q values are generally stored in a matrix called the Q-Table. The algorithm uses a temporal-difference approach to iteratively update the values of this table after each interaction, thanks to the modified Bellman equation:

$$Q(s, a) \leftarrow Q(s, a) + \eta\Big(r + \kappa \max_{a'} Q(s', a') - Q(s, a)\Big) \tag{35}$$

Remark 7.5. *The Q-Learning hyperparameters are denoted η (learning rate) and κ (discount factor) in order to avoid any confusion with the exponents $\alpha, \beta, \delta, \gamma$ of the gravity law (Definition 3.3), which the agent precisely seeks to calibrate in Section 7.5.*

This update formula relies on several hyperparameters essential to the agent's proper learning:

— The learning rate ($\eta \in [0, 1]$): It determines the speed at which new information overwrites old information. An $\eta = 0$ prevents the agent from learning, while an $\eta = 1$ makes it totally amnesic with respect to its past experience, keeping only the very last observation.

— The discount factor ($\kappa \in [0, 1]$): It weights the importance of future rewards relative to immediate rewards. A κ close to 0 will make the agent myopic and opportunistic, while a κ close to 1 will push it to favor long-term strategies.

— The learning target ($r + \kappa \max_{a'} Q(s', a')$): It represents the real observed value, that is, the immediate reward r added to the estimate of the best possible value in the new state s'.

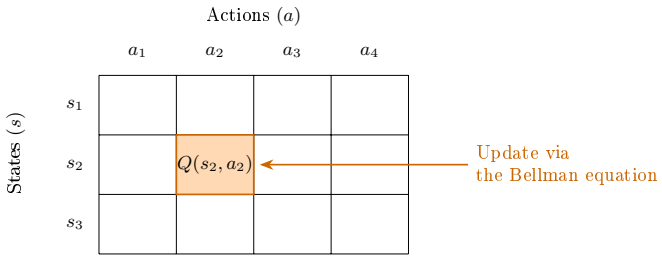


FIGURE 27 – Update mechanism of a Q-Table cell.

7.5.3 The Exploration vs Exploitation dilemma

To converge toward the optimal policy, the agent must traverse its environment exhaustively. It faces a permanent compromise:

— Exploitation: Choose the action with the highest known Q value in order to maximize the immediate gain.

— Exploration: Choose a sub-optimal or random action with the aim of discovering new trajectories that are potentially more profitable in the future.

To manage this dilemma, the most common approach is the ϵ-greedy strategy. At each step, the agent chooses to explore with probability ϵ, and to exploit its knowledge with probability $1 - \epsilon$. In practice, the value of ϵ is often initialized at 1 (total exploration at the beginning of learning) then decreases progressively toward 0 (maximal exploitation once the environment is known).

7.5.4 Application to the functional model of health-state quantification

Why this synergy is relevant: The functional model developed introduces the operator $\zeta(t)$, computing a normalized scalar in the interval $(-1, 1)$. This mathematical formalism natively solves one of the recurrent problems of reinforcement learning: the explosion or instability of the reward function. By directly using $\zeta(t)$ as the reward signal, the agent has a stable, continuous and bounded performance metric.

Mathematical formalization of the coupling: To apply Q-Learning to this dynamic environment, the concepts of the article are projected into a Markov Decision Process (MDP):

— State Space ($\mathcal{S}$): At instant t, the state s_t is defined by the triplet $\{\mathcal{E}(t), \mathbf{W}(t), \zeta(t)\}$, where $\mathcal{E}(t)$ represents the set of active events and their 5 quantifiable attributes (residual gravity, etc.), and $\mathbf{W}(t)$ is the interaction matrix.

— Action Space ($\mathcal{A}$): The action a_t corresponds to the regulation levers of the operator, such as the forced injection of a compensatory event (neutralization), the modification of a gravity attribute, or the structural modification of an interdependence relation in $\mathbf{W}(t)$ (for example, the attenuation of an amplification effect).

— Reward Function (r_t): The immediate reward returned to the agent is directly correlated with the evolution of the health of the system. Two approaches are conceivable:

1. Absolute reward: $r_t = \zeta(t)$, encouraging the agent to keep the system as close as possible to the healthy state (+1).
2. Differential reward: $r_t = \zeta(t) - \zeta(t-1)$, rewarding any action leading to a trajectory of clinical improvement of the environment.

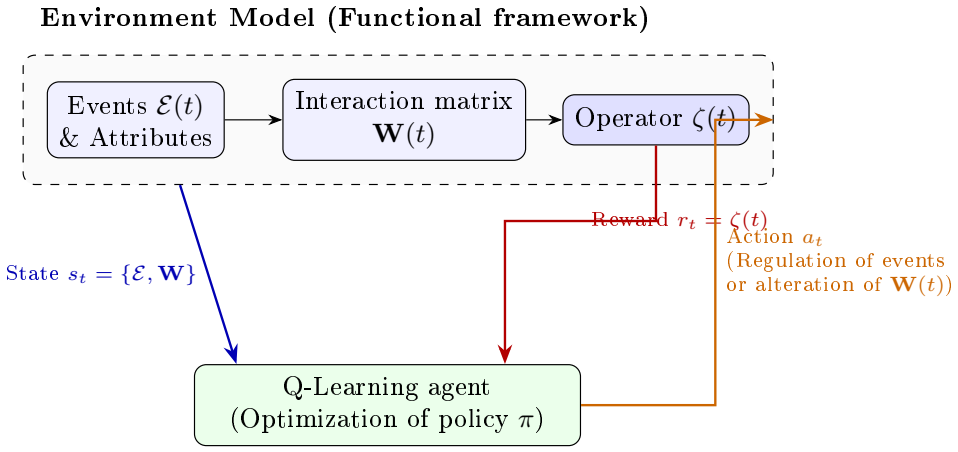


FIGURE 28 – Integration architecture of Q-Learning on the functional model of health-state quantification.

Thanks to this architecture, the Q-Learning algorithm learns empirically the best policy for managing environmental crises. Faced with a shock (unforeseen event of high gravity), the agent will select the action that minimizes the drop of $\zeta(t)$ in the long term, taking into account the amplification or feedback phenomena encoded in the matrix $\mathbf{W}(t)$.

7.5.5 Analysis of the results

This subsection is devoted to the critical evaluation of the performance of this approach. Through the joint examination of the convergence trajectory of the MAPE, of the structure of the aggregated Q-table and of the distribution of state visits, we will analyze the agent's ability to efficiently explore the solution space and to stabilize the values of the parameters α, β, γ and δ.

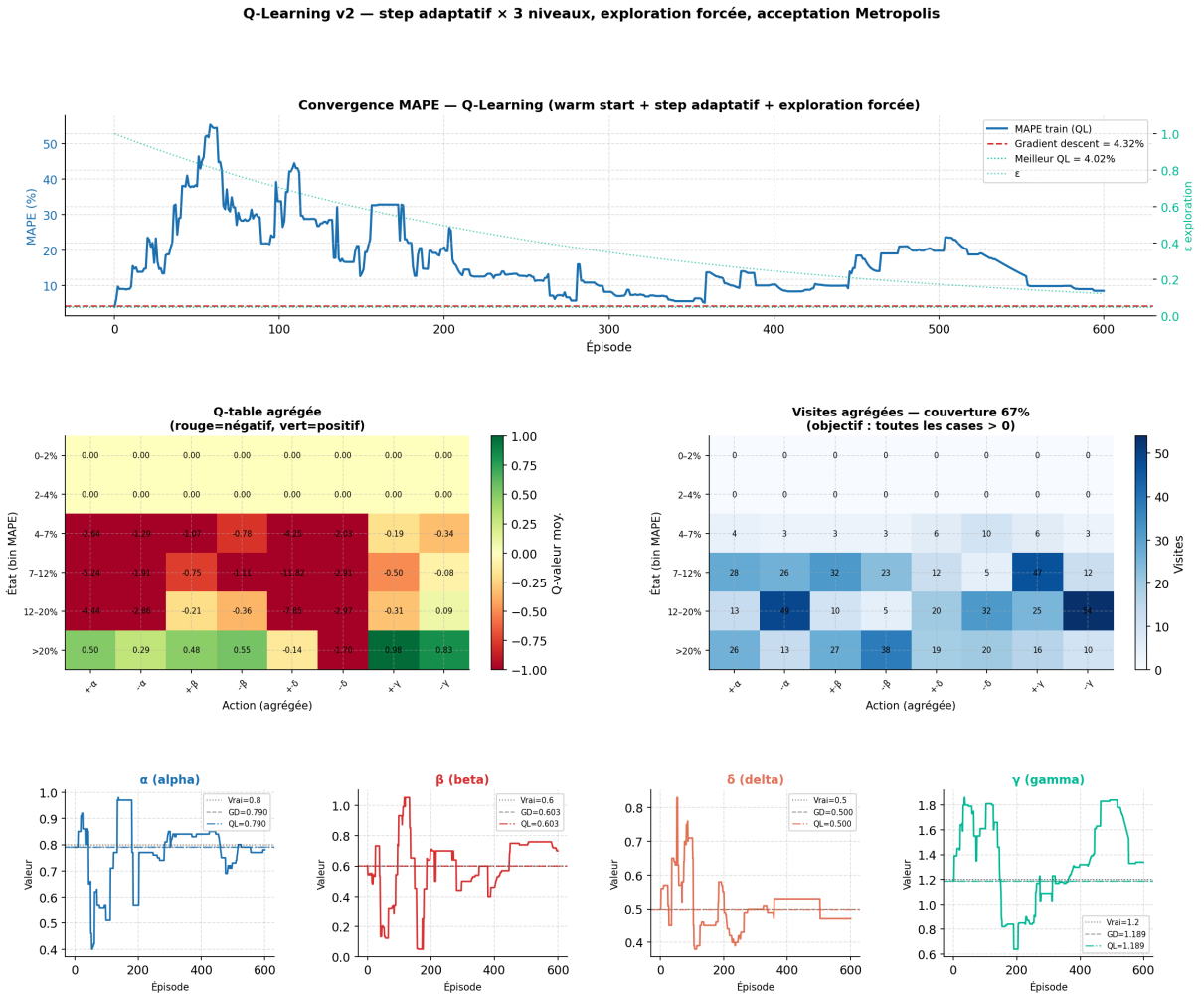


FIGURE 29 – Compiled information related to Q-Learning

— Evolution of the performances and of the coefficients (first graph): The maximal score reached so far is observed around 350 episodes. The analysis of the coefficients at this stage is entirely consistent with this result: around the 350th episode, the parameters β, δ and γ have almost recovered their initial values (those that led to the best historical score), with the exception of the parameter α which lies very slightly above.

— Frequency of states and reward table: No cell has been visited among those associated with a MAPE between 0 % and 4 %. This directly explains the zero scores (0) observed within the reward table located on the left.

These elements highlight that at the beginning of the process, the correction of the MAPE is relatively easy. Since it is initially high (and therefore poorly performing), the probability of converging toward combinations of parameters that improve it is markedly higher.

— Analysis of the impact of the parameter δ: Whatever the interaction with δ, the score remains systematically negative despite a relatively large volume of visits. It would therefore seem that the value of δ negatively impacts the MAPE. This would tend to show a structural weakness of the model itself, or at least of the way the parameter δ is currently integrated into it.

TABLE 4 – Ratio between the mean Q-value and the number of visits (Q-value/Visits)

State (MAPE bin)	$+\alpha$	$-\alpha$	$+\beta$	$-\beta$	$+\delta$	$-\delta$	$+\gamma$	$-\gamma$
0–2%	–	–	–	–	–	–	–	–
2–4%	–	–	–	–	–	–	–	–
4–7%	−0.6600	−0.4000	−0.3567	−0.2600	−0.7083	−0.2030	−0.0317	−0.1133
7–12%	−0.1871	−0.0735	−0.0234	−0.0483	−0.9850	−0.5820	−0.0106	−0.0067
12–20%	−0.3415	−0.0584	−0.0210	−0.0720	−0.3925	−0.0928	−0.0124	0.0017
>20%	0.0192	0.0223	0.0178	0.0145	−0.0074	−0.0850	0.0612	0.0830

The analysis of this ratio table allows us to draw the following conclusions:

— Cost of exploration in the high-precision zone: The more the algorithm progresses toward a low MAPE (high precision), the more penalizing the state visits become (more marked negative ratios). Each action undertaken in these critical zones degrades the mean Q-value more strongly, which reflects a difficulty of the agent in stabilizing its gains.

— Action granularity problem: It would seem that the size of the modification steps of the hyperparameters $(\alpha, \beta, \gamma, \delta)$ is no longer fine enough to allow the agent to access the higher performance levels (the MAPE ranges between 0 % and 4 %, which remained unexplored).

— Limiting factors: This convergence defect can be attributed to two main causes:

1. An excessively fast exponential decay of the exploration parameter ϵ, which freezes the agent's policy before it has been able to refine its choices.
2. The discretization effect (the splitting of the MAPE into "bins"), whose coarseness masks the fine gradients and prevents the algorithm from perceiving micro-improvements.

In-depth analysis of the discretization effect The splitting of the MAPE into discrete intervals (or "bins") introduces a classical challenge in reinforcement learning: the state aggregation problem. In the high-precision zones, this mechanism limits the convergence of the algorithm through two major phenomena:

— The "Blind Spot" phenomenon: The MAPE is a continuous variable, whereas the Q-table relies on discrete states. By grouping performances into intervals, the variations within the same state become invisible to the agent. For example, if the agent goes from a MAPE of 11.9 % to 7.1 %, the real improvement is massive (≈ 5 %). Yet, for the algorithm, the starting state and the arrival state remain identical (the 7–12 % interval). Without a reward function that is extremely sensitive to the underlying continuous variations, the Bellman equation perceives no progression, making the agent mathematically "blind" to its own successes.

— The edge effect: The discretization creates an artificial distortion of the perception of distances. A tiny variation of 0.2 % (going from 12.1 % to 11.9 %) causes a complete change of state (12–20 % → 7–12 %) and is perceived as a major transition. Conversely, a significant optimization effort of 4 % (going from 11.5 % to 7.5 %) leaves the agent in the same state, without any gratification. This inconsistency disturbs the update of the Q-table values in the zones close to the optimum, precisely where the precision of the actions is crucial.

Remark 7.6 (Why not simply reduce the intervals?)**.** *Reducing the width of the MAPE bins is indeed the first lever that comes to mind, but it runs into a sampling compromise: with only a few hundred episodes (Figure 29), a finer discretization multiplies the number of states $s \in \mathcal{S}$ without guaranteeing a sufficient number of visits per state, which slows down the convergence of the Bellman equation (35) accordingly and may, in practice, degrade the stability of learning rather than improve it. Two more robust avenues are conceivable: a progressive refinement of the bins over the course of training (on the same principle as the adaptive refinement of nodes of Section 7.3.3), or — a more radical solution, already identified as a perspective (Section 9) — the pure and simple replacement of the tabular Q-table by a continuous approximator of $Q(s, a)$.*

7.6 Adoption of a new event

Until now, we have assumed that each event belonged explicitly to an identified cluster. This hypothesis is however restrictive, because in practice new events may appear without direct correspondence with an already observed category.

Principle of hybrid representation.
We propose to represent an unknown event as a lexical combination of existing clusters. The idea is that a new event can be reconstructed from the structures already learned in the space of events, according to their conceptual proximity.

For example, a tsunami can be interpreted as a combination of an earthquake and a flood.

Law of lexical resemblance.
We have previously introduced a similarity function defined between two events that quantifies their lexical proximity in the space of events:

$$\mathrm{Sim}(e_i, e_j) \in [0, 1].$$

From this quantity, we define an affinity measure between an event and a cluster as an aggregation of the internal similarities:

$$\mathrm{Sim}(e, C_k) = \frac{1}{|C_k|} \sum_{e_j \in C_k} \mathrm{Sim}(e, e_j). \tag{36}$$

We denote by π_k these decomposition weights, in order to avoid any confusion with the static weights ω_i of the events (Definition 6.3): they are then obtained by normalization:

$$\pi_k = \frac{\mathrm{Sim}(e_{\mathrm{new}}, C_k)}{\sum_{j=1}^{K} \mathrm{Sim}(e_{\mathrm{new}}, C_j)} \qquad \text{with} \qquad \sum_{k=1}^{K} \pi_k = 1. \tag{37}$$

Thus, the decomposition is written:

$$e_{\mathrm{new}} \approx \sum_{k=1}^{K} \pi_k \, C_k. \tag{38}$$

Example.
For an event of the tsunami type, we typically obtain:

$$e_{\mathrm{tsunami}} \approx 0.6 \, C_{\mathrm{earthquake}} + 0.4 \, C_{\mathrm{flood}}. \tag{39}$$

Propagation of gravity.
In this framework, the gravity associated with a composite event is defined by weighted combination:

$$\Gamma(e_{\mathrm{new}}) = \sum_{k=1}^{K} \pi_k \, \Gamma_k, \tag{40}$$

where Γ_k represents the gravity associated with the cluster C_k.

This formulation makes it possible to naturally generalize the model to unknown events, while keeping the structure learned in the space of clusters.

8 Model synthesis

Object	Definition	Space	Origin	Ref.
Event e_i	(D, F, R, P)	$\mathbb{R}^4$	Author	Def. 2.3
Gravity $\mathcal{G}_i(t)$	$\mathcal{D}_{\text{eff}}^{\alpha}\mathcal{F}^{\beta}\mathcal{R}^{\delta}/(1+\mathcal{P}^{\gamma})$	$\mathbb{R}^+$	Author	Def. 3.4
Matrix $\mathbf{W}(t)$	$[W_{ij}(t)]$	$\mathbb{R}^{N\times N}$	Author	Def. 4.2
Eff. gravity $\widetilde{\mathcal{G}}_i$	$(\mathbf{I}+\mathbf{W})\mathbf{G}$	$\mathbb{R}$	Author	Def. 4.3
Weight ω_i (static)	$\in [-\omega_{\max}, \omega_{\max}]$	$\mathbb{R}$	Author	Def. 6.3
Sheet f^*	$\arg\min \bar{I}(f)$	$C^{\infty}(\Omega)$	Author	Def. 6.8
Norm. weight $\bar{\omega}_i$	$\omega_i/\|\omega\|_1$	$\mathbb{R}$	Author	Def. 6.4
Health state ζ_λ	$\tanh\left(\frac{1}{\lambda}\sum_i \bar{\omega}_i\widetilde{\mathcal{G}}_i\right)$	$(-1,1)$	Author	Eq. (24)

TABLE 5 – Synthesis of the mathematical objects. All the elements are original contributions of the author unless otherwise stated.

Transformation chain:

$$\{e_i\} \xrightarrow{\Gamma_{\hat{\theta}_{k(i)}}(t)} \mathbf{G}(t) \xrightarrow{\mathbf{I}+\mathbf{W}(t)} \widetilde{\mathcal{G}}(t) \xrightarrow{\langle\bar{\omega},\cdot\rangle} S(t) \xrightarrow{\tanh(\cdot/\lambda)} \zeta_\lambda(\mathcal{E}(t)) \in (-1,1). \tag{41}$$

9 Discussion and conclusion

9.1 Synthesis of the contributions

This work proposed a unified mathematical framework for the quantitative modeling of a dynamic environment. Three contributions structure the whole construction.

First, the event-based representation of the environment (founded on five quantifiable attributes and a characterization operator Γ directly integrating the temporal dynamics via the residual gravity $\mathcal{D}_{\text{eff}}(t)$) makes it possible to dispense with an explicit modeling of the spatial frame, while remaining applicable to fields as varied as finance, ecology or risk management.

Second, the interaction matrix $\mathbf{W}(t)$ formalizes the cross effects between coexisting events (amplification, feedback, neutralization), a dimension often neglected in purely additive approaches.

Third, the operator ζ_λ, based on tanh and enriched by an adaptive dilation mechanism, synthesizes all of this information into a bounded and interpretable indicator, whose construction (L^1-normalized static weights, variational sheet for the initialization of new events, smoothing by adaptive splines) was validated by a series of preliminary experiments reaching a MAPE of 3.35% to 4.32% depending on the configuration tested, for a coefficient of determination $R^2 = 0.9913$.

9.2 Current limits of the model

Several limitations, already mentioned occasionally throughout the text, deserve to be gathered and discussed more systematically.

— Subjectivity of the frame, of the order of magnitude and of the initial weights. The frame C (Definition 2.2), the order of magnitude h (Remarks 2.3 and 2.4) and the initial weights ω_i (Definition 6.3) remain largely exogenous choices of the model. The variational sheet (Section 6.6) alleviates this problem for the weights of new events by interpolating the observed weights, but does not remove the initial dependence on an already annotated event base, nor the need for the user to fix C and h a priori.

— Sensitivity of lexical clustering. The Jaccard similarity approach (Proposition 3.1) relies on a threshold τ whose choice remains empirical: a threshold that is too low merges families of events with distinct gravity laws, whereas a threshold that is too high artificially fragments homogeneous clusters.

— Quadratic complexity of the interaction matrix. As indicated in the Remark following Algorithm 1, the computation of the effective gravity $\widetilde{\mathcal{G}}(t) = (\mathbf{I} + \mathbf{W}(t))\mathbf{G}(t)$ is $O(N(t)^2)$ in its naive implementation. For environments with a very large number of active events, an exploitation of the sparsity of $\mathbf{W}(t)$ (most pairs of events not interacting) would be necessary to guarantee scaling.

— Discretization and state aggregation in Q-Learning. The detailed analysis in Section 7.5 highlighted the "blind spot" phenomenon: the splitting of the MAPE into discrete intervals masks the fine gradients and limits the agent's ability to distinguish real improvements in the high-precision zones (MAPE $< 4\,\%$), without a simple reduction of the interval width being sufficient to solve the problem (Remark 7.6). The replacement of the tabular Q-table by a continuous approximation of $Q(s, a)$ (for example a neural network [1], in the manner of Deep Q-Learning) would constitute a more robust answer to this limitation.

— Asymmetric loss function not yet integrated into the calibration. The asymmetry criterion introduced in Section 6 (Equation (23)) remains, at this stage, a conceptual framework: the per-cluster calibration (Definition 3.7) continues to rely on the symmetric MAPE. The explicit integration of $\ell(\hat{g}, g)$ into the optimization procedure (in particular for events with strong negative impact, where $c_- > c_+$ — see Remark 6.6 for avenues to calibrate this ratio) would make it possible to penalize underestimations more severely and to align the calibration with the operational stakes of risk management mentioned in the introduction.

— Absence of validation on real data. All the results presented (Sections 6 and 7) rely on a randomly generated event base, structured around three synthetic clusters (financial, climate, health). While this approach makes it possible to validate the internal consistency of the model and the stability of its calibration, a confrontation with corpora of real events, annotated by experts or extracted from historical databases, remains necessary to assess the practical scope of the proposed framework.

9.3 Perspectives

Several research directions follow naturally from the limits identified above.

In the short term, the automation of the choice of the lexical similarity threshold τ, the automation of the computation of the characteristic order of magnitude h from the frame C (Remark 2.4), and the integration of the asymmetric loss into the per-cluster calibration appear as the most directly accessible extensions.

In the medium term, the replacement of the discrete Q-table by a continuous approximator would make it possible to handle environments of realistic size.

Finally, in the longer term, the validation of the model on real datasets, coupled with an automation of the operator Ψ by a natural language processing model (Definition 3.1), would open the way to an operational system for the continuous monitoring of the health state of an environment, whether financial, ecological or organizational.

9.4 Conclusion

Ultimately, this work proposes a coherent and modular mathematical formalization of a dynamic environment, from the individual characterization of events to their synthesis into a single bounded indicator.

The frame–events–time axis hierarchy, the residual gravity operator Γ, the interaction matrix $\mathbf{W}(t)$ and the health state operator ζ_λ constitute the building blocks of an extensible framework, whose relevance has been supported by encouraging preliminary experimental results. The limits identified, the subjectivity of certain initial choices, the quadratic cost of interactions, the state aggregation in reinforcement learning, do not call into question the validity of the formal framework, but circumscribe its current scope of application and trace the axes of a future development.

Annexes

A.1 Properties of the functional form of Γ (Definition 3.3)

Proof. (1) Monotonicity: $\partial G/\partial \mathcal{D} = \alpha \mathcal{D}^{\alpha-1}\mathcal{F}^{\beta}\mathcal{R}^{\delta}/(1+\mathcal{P}^{\gamma}) > 0$, likewise for $\mathcal{F}$ and $\mathcal{R}$. (2) Decrease in $\mathcal{P}$: the partial derivative is ≤ 0 (strictly for $\mathcal{P} > 0$). (3) Positivity: all the factors are positive. (4) $\lim_{\mathcal{P}\to+\infty} G = 0$, $\lim_{\mathcal{P}\to 0} G = \mathcal{D}^{\alpha}\mathcal{F}^{\beta}\mathcal{R}^{\delta}$. □

A.2 Proof of Proposition 4.1

Proof. Adding $W_{ii} = 0$ in the sum of Definition 4.3: $\widetilde{\mathcal{G}}_i = \sum_j (\delta_{ij} + W_{ij})\mathcal{G}_j$, i.e. $\widetilde{\mathcal{G}} = (\mathbf{I} + \mathbf{W})\mathbf{G}$ in matrix notation. □

A.3 Proof of Proposition 6.1

Proof. (i) $|\tanh(x)| < 1$ because $e^x + e^{-x} > |e^x - e^{-x}|$, image $(-1, 1)$. (ii) $\tanh(x) > 0 \iff x > 0$ (sign preservation). (iii) $\tanh'(x) > 0$ (strictly increasing, injective); surjectivity by the intermediate value theorem. □

A.4 Proof of Proposition 6.2

Proof. (i) For $c > 0$, $\bar{\omega}_i = c\omega_i/(c\|\omega\|_1)$ unchanged. (ii) $\omega_{N+1} = 0$ does not affect $\|\omega\|_1$. (iii) $|S| \leq \sum_i |\bar{\omega}_i||\widetilde{\mathcal{G}}_i| \leq \max_i |\widetilde{\mathcal{G}}_i|$ because $\sum_i |\bar{\omega}_i| = 1$. □

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